KU LEUVEN

Chapter 5. Postprocessing

Data Science for Business

D0I74A

Prof. dr. Wouter Verbeke

Prof. dr. Bart Baesens



Course schedule

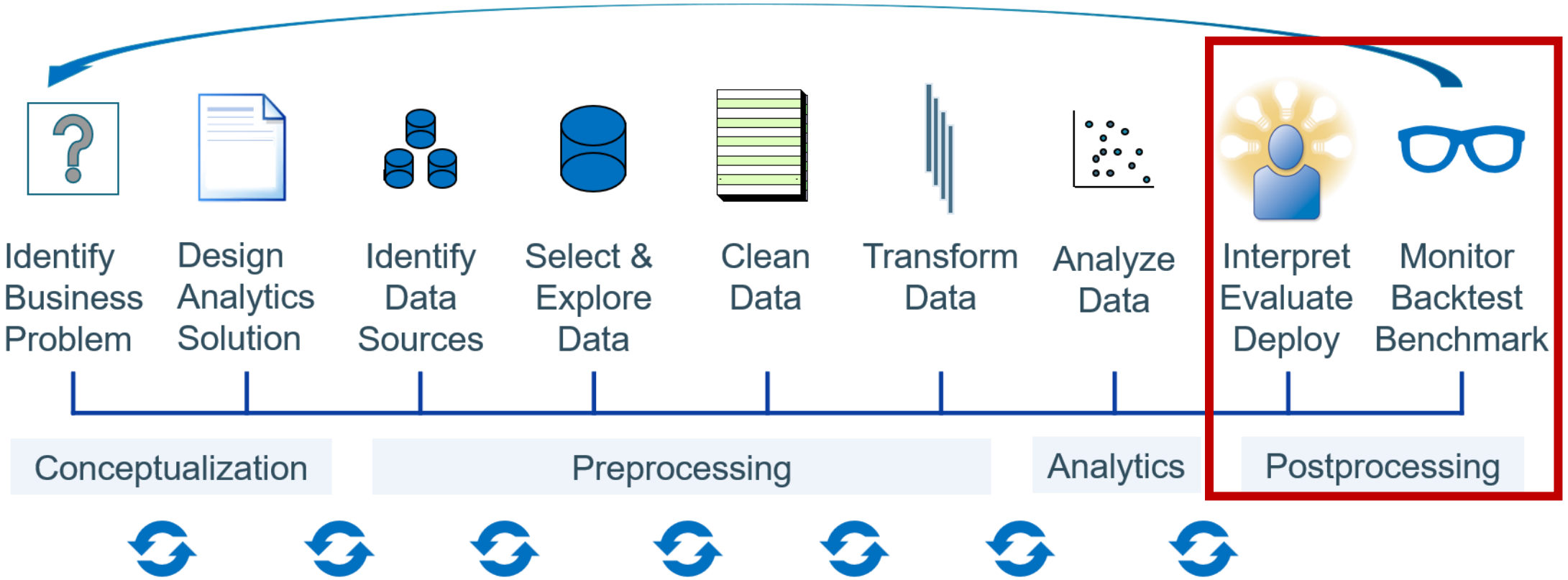
BLOCK1

1	21/09/2026	Chapter 0. Course Outline Chapter 1. Data Science for Business	
1	22/09/2026	Chapter 1. Data Science for Business (ctd.)	
		Hands-on #1: Building a first pipeline	**Homework 1**
2	28/09/2026	Chapter 1. Data Science for Business (ctd.) Chapter 2. Exploratory Data Analysis & Preprocessing	
2	29/09/2026	Chapter 2. Exploratory Data Analysis & Preprocessing (ctd.)	
		Hands-on #2: EDA & Preprocessing	**Homework 2**
3	05/10/2026	Chapter 2. Exploratory Data Analysis & Preprocessing (ctd.)	
3	06/10/2026	Chapter 3. Predictive Analytics - Decision Trees	
4	12/10/2026	Chapter 4. Predictive Analytics - Nearest Neighbors & Ensembles	
4	13/10/2026	Chapter 5. Postprocessing	
5	19/10/2026	Chapter 5. Postprocessing	
		Assignment 1	**Report 1: 10/11**

Data science process model

Chapter 5

- Practical steps in elaborating a customer churn prediction model:



Outline

1. Evaluation methodology
2. Evaluation measures for binary classification
3. Evaluation measures for multiclass classification
4. Evaluation measures for regression
5. Evaluation beyond predictive accuracy
6. Case: Credit risk modeling
   - Scorecards
   - Ratings and calibration
   - Backtesting and benchmarking
7. XAI

Learning objectives

- At the end of this chapter, students should be able to:
  - Understand, explain, select, apply and motivate the various **evaluation methodologies** discussed in this chapter
  - Explain and interpret the various **evaluation measures** for model performance assessment
  - Discuss evaluation and postprocessing **beyond assessment of predictive power**

1. Evaluation methodology

Evaluation methodology

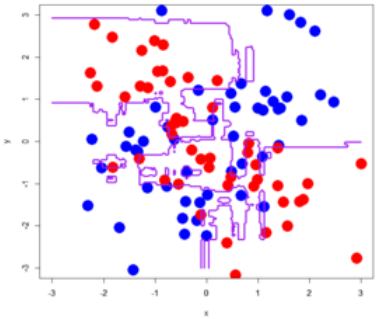
- Methodology?

- Why needed for evaluation?
  - Simply looking at accuracy or errors?
  - Simply …

- Various approaches:
  - Split sample
  - Cross-validation
  - Single sample

Split sample method

Perfectly fitting
the training data
= overfitting
(cf. Chapter 3)



- Data scientist's approach
  - We want to know how well the model will **predict** when used in the future for decision-making
    - I.e., when applied on new, unseen observations = **not used for developing the model = generalization power**
  - Why is that important?
    - The model **IS** the data used for developing it! (*data* transformed into *information*)
    - Therefore, it will predict well (to a certain extent) the observations used for developing it
    - We can *easily* develop a model that *perfectly fits* the *training data*
  - We therefore **evaluate (or *simulate*) performance** on new, unseen observations
    - E.g., by **splitting the sample** prior to developing the model in a **training and test set**
      - The model is developed only using the training set
      - The model is evaluated on the test set … which was not used to develop the model
  - Pragmatic approach
  - Applicable **if large data set**
    - Large?
    - More than 1000 observations

With kNN (cf. Chapter 4),
the data literally **is** the model

Split sample method

BEWARE: DATA LEAKAGE!

- How to split?
  - Typically:
    - Training set: randomly selected 70% of the observations (or 2/3, or ..)
    - Test set: randomly selected 30% of the observations (or 1/3, or ..)
    - **Why random?**
  - **Out-of-sample** evaluation

- **Stratified sampling**: random sampling, ensuring the *distribution* of the target variable is the same in both training and test set
  - Why?
  - Stratifying predictor variables? (cf. Chapter uplift modeling: matching)

Data leakage!

A simple rule of thumb: *"At the exact moment I need to make this prediction, would I actually have this piece of information?"* If not, it's leakage.

A warning sign: results that look too good to be true

- **Leakage**:
  - The use of information in the model training process
  - Which would not be expected to be available at prediction time
  - Causing the predictive scores to overestimate the model's utility
  - When the model would be run in a production environment
- Often subtle and indirect, making it sometimes hard to detect and eliminate
- **Target leakage:**
  - Information on the target included in the predictors
    - Hospital readmission model uses "discharge medication prescribed," which is only recorded at the end of stays.
  - Information that only exists afterwards
    - Variable 'Exit-talk executed' in HR turnover prediction dataset
    - A default model includes "number of collection letters sent"

https://en.wikipedia.org/wiki/Leakage_(machine_learning)
https://towardsdatascience.com/data-leakage-in-machine-learning

Data leakage!

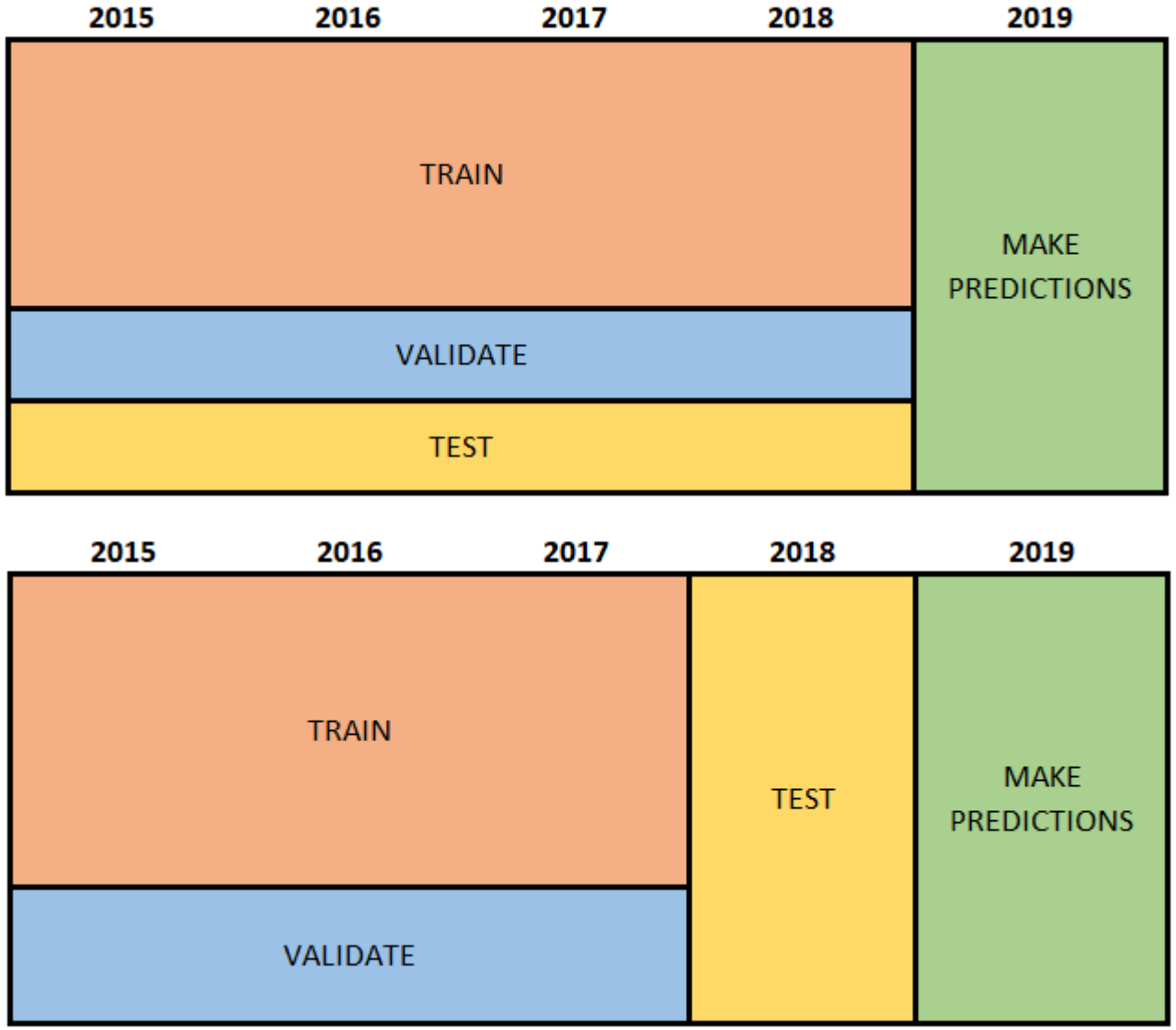
A simple rule of thumb: *"At the exact moment I need to make this prediction, would I actually have this piece of information?"* If not, it's leakage.

A warning sign: results that look too good to be true

- **Test data leakage:**
  - Preprocessing with the whole dataset
    - E.g., WOE calculation before train-test split
  - Time travel in the data split
    - E.g., train on 2025 and 2026 data, and test on random 2025 days, fraud patterns from 2026 were unknown at that moment.
  - The same unit on both sides of the split
    - Customer has 20 transactions, and some land in training while others land in test. The model recognises the customer instead of learning the pattern, so it does worse on genuinely new customers.
  - Tuning on the test set
    - You try 50 model versions and keep the one with the best test-set score.
    - The test set has effectively become training data, with reported performance too optimistic

Split sample method

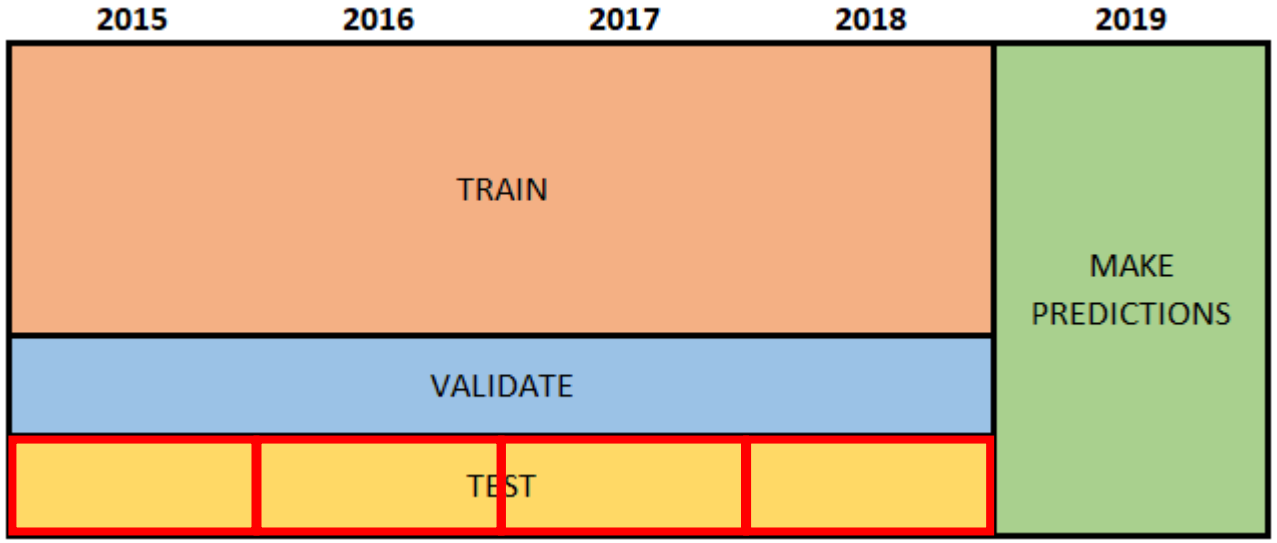
- **Out-of-sample**
- Alternatively: **out-of-time** evaluation
- Why?
  - Handling the time-dimension!
  - Avoiding test set leakage
  - Ensuring relations hold in time
  - Examples when no time-dimension?
- But:
  - Most recent data not used for developing, only for evaluation
  - If the most recent data is the most representative data, then …



https://towardsdatascience.com/why-isnt-out-of-time-validation-more-ubiquitous-7397098c4ab6

Split sample method

- Out-of-sample with inspection of performance across time blocks in test-sample
  - In each time block, check performance: stable across time?

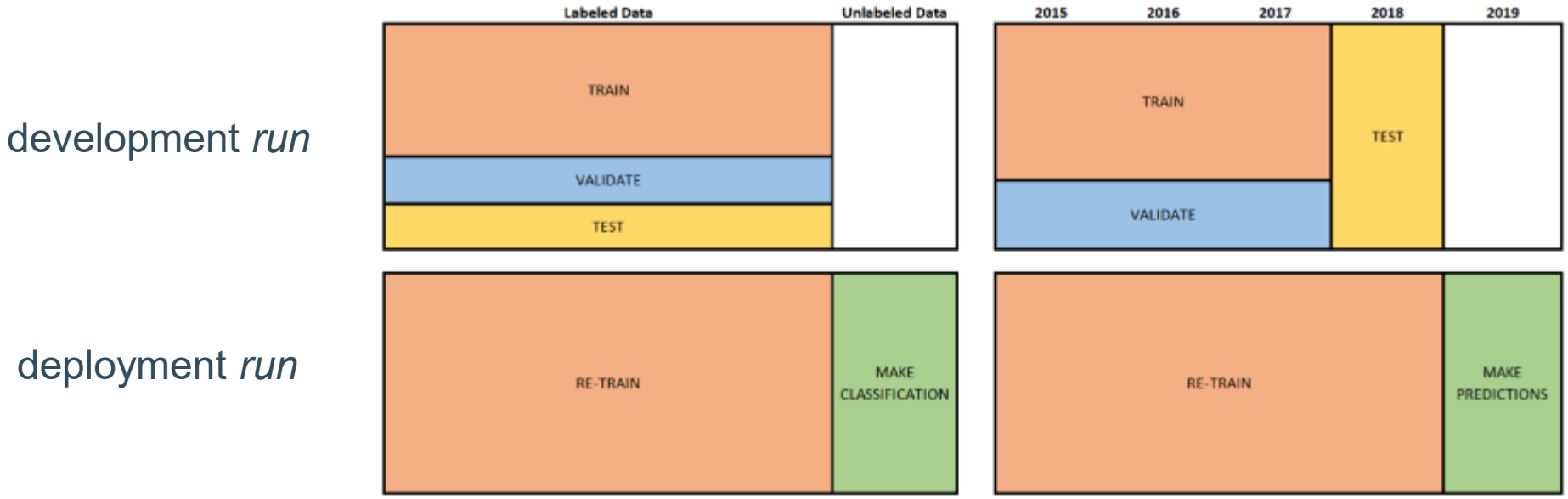


https://towardsdatascience.com/why-isnt-out-of-time-validation-more-ubiquitous-7397098c4ab6

Split sample method

Depending on the method, you may compare the models from the development and deployment runs

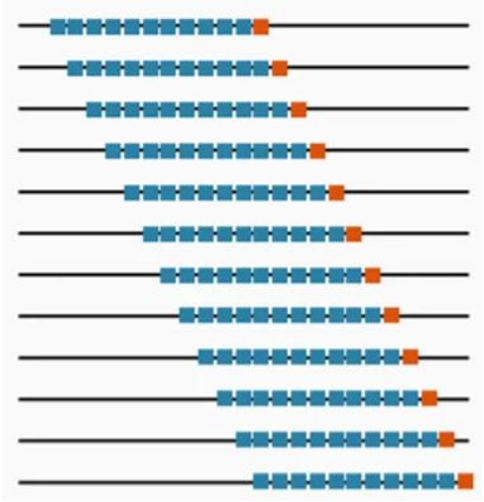
- Maximizing performance: using all data to train final model
  - Performance of development run model *representative* for performance of **deployment run** model?

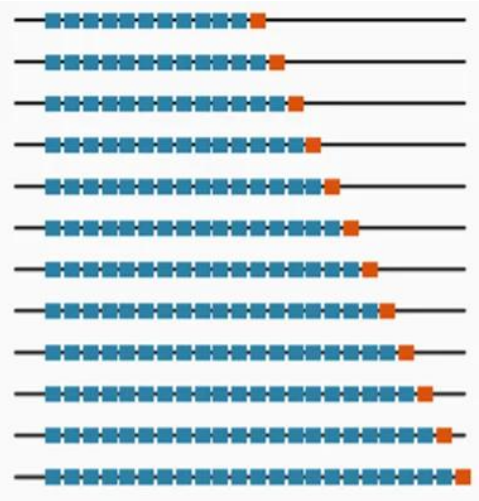


https://towardsdatascience.com/why-isnt-out-of-time-validation-more-ubiquitous-7397098c4ab6

Split sample method

- **Rolling window evaluation**
  - Cf. Chapter 7. Forecasting
  - = iterative out-of-time evaluation
  - ‘Blue data’ either available for
    - (re-)training in subsequent time windows
    - latest available information for making predictions for next window
- **Expanding window evaluation**
- **Optimal window** for training?





https://towardsdatascience.com/why-isnt-out-of-time-validation-more-ubiquitous-7397098c4ab6

Split sample method

- **Optimal window** for training?
  - **Tune** the training window



https://towardsdatascience.com/why-isnt-out-of-time-validation-more-ubiquitous-7397098c4ab6

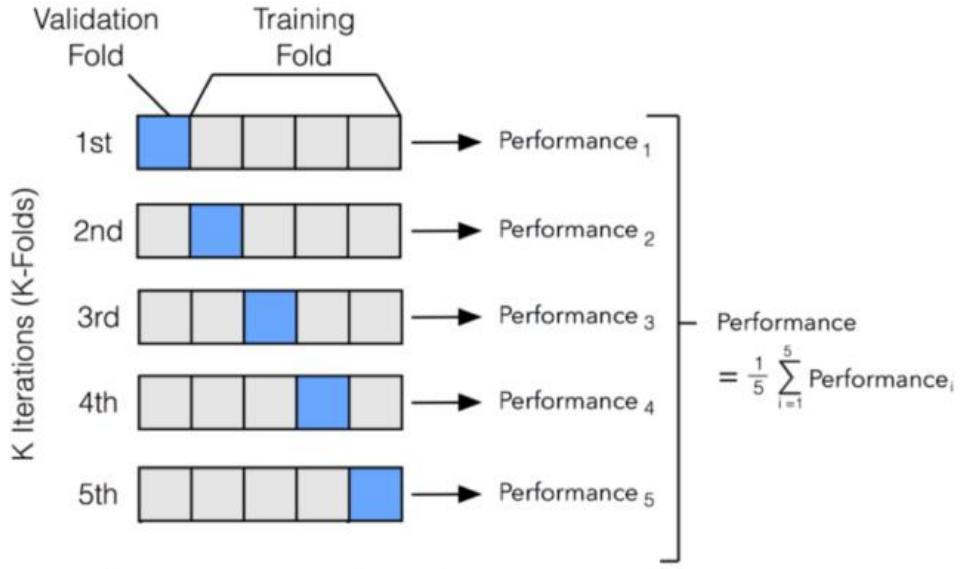
Split sample method

- Random split-up: **out-of-sample** evaluation
- Alternatively:
  - **Out-of-time** evaluation
  - **Out-of-universe** evaluation
    - Evaluated on different (sample of a) “population” than the (sample of a) “population” on which the model was developed (cf. test set leakage)
    - Various *degrees* of ‘out-of-universe’
      - Different customers
      - Different customer segment
      - Different service or product
      - …

Cross-validation

Validation fold or test fold?
Nested cross-validation!

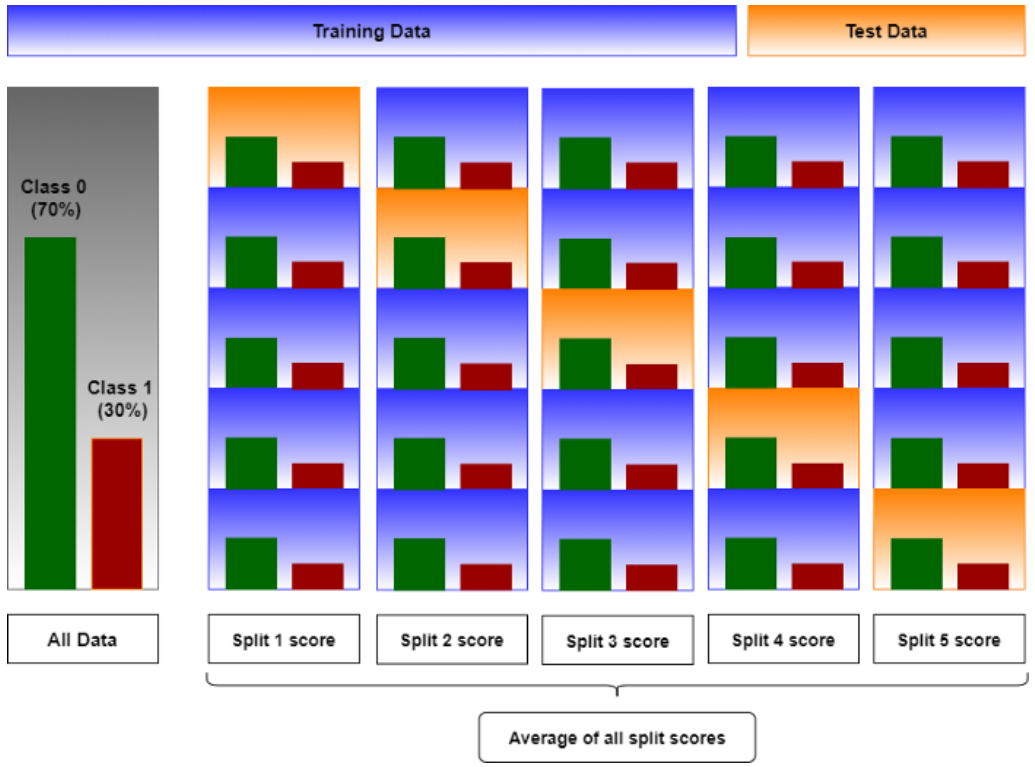
- Adopting the split sample method: **30% of the data is not used for developing the model**
- If a small dataset (small: < 1000 observations):
  - More **efficient** method? (efficient in using the data optimally for maximizing performance)
  - More **reliable** method? (since performance evaluated on a small sample may not be representative)
- **N-fold cross-validation** (or better: **N-fold cross-testing**)
  - Randomly split in n folds
  - N times develop a model using n-1 folds, and evaluate on hold-out fold
  - Final model:
    - Can be randomly selected (or: retrain on all data)
    - Do inspect the variation across the n models
      - Decision trees versus logistic regression!
  - How much data used in each development?
    - (n-1)/n * 100%
    - E.g.: 10-fold cross-validation: 90%



https://medium.com/analytics-vidhya/cross-validation-techniques-bacb582097bc
https://scikit-learn.org/stable/auto_examples/model_selection/plot_nested_cross_validation_iris.html

Cross-validation

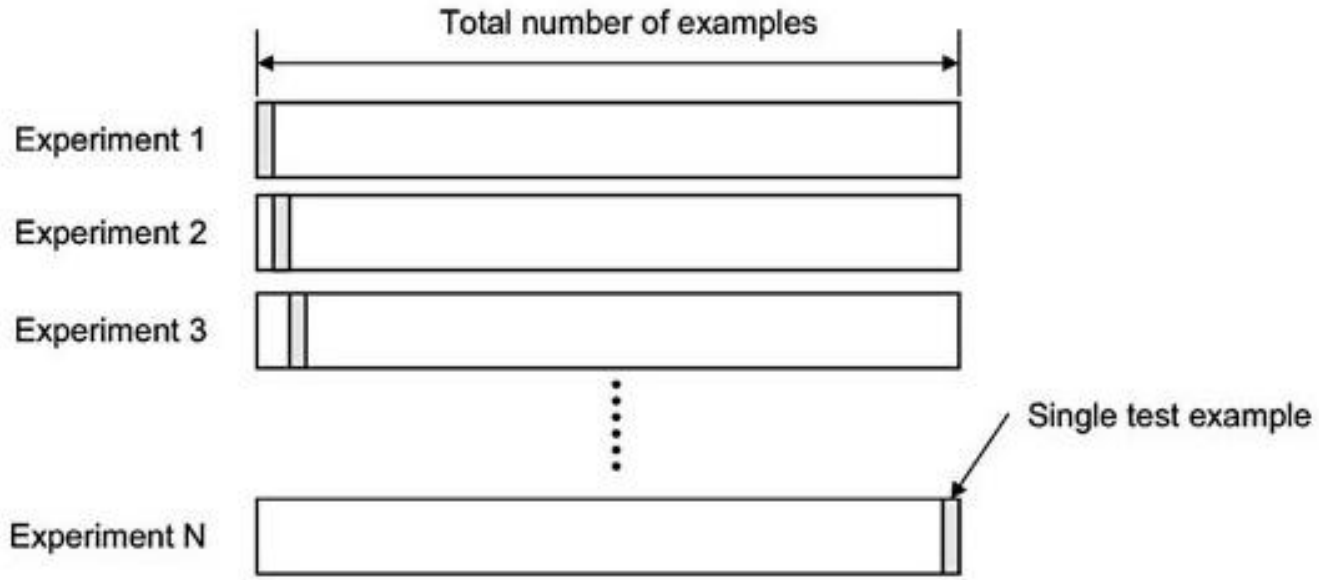
- **Stratified** n-fold cross-validation
  - Important when **imbalanced class distribution**!



https://medium.com/analytics-vidhya/cross-validation-techniques-bacb582097bc

Cross-validation

- **Leave-one-out cross validation**
  - +: Minimal variability across the resulting models
  - -: Computational requirements



https://medium.com/analytics-vidhya/cross-validation-techniques-bacb582097bc

Single sample method

"Everything should be made as simple as possible, but not simpler"

It can scarcely be denied that the supreme goal of all theory is to make the irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience.

- **Statistical approach** (vs. 'data scientist's approach')
  - To explain
  - What proportion of the observed variability in the dependent variable is explained by the independent variables in the model?
  - No test set: model evaluated on the same data as used to evaluate the model
    - Overfitting?!
    - More complexity obviously will yield smaller errors
  - Therefore:
    - Performance = f(training error, model complexity)
    - Reward for reduced training error
    - Punish for model complexity
    - **Model complexity** can be measured by **the number of estimated parameters**

Single sample method

L = Likelihood: see also Logistic regression, ANN, …

- Akaike Information Criterion (AIC)

  AIC = -2 log **L** + 2 (nr of parameters)

- Bayesian Information Criterion (BIC)

  BIC = -2 log **L** + (nr of parameters) * log(nr of obs)

  (also known as Schwarz Bayesian Criterion (SBC))

- Best model has minimum AIC or BIC

- Both criteria model the **trade-off between the fit of the model and its complexity** (BIC is punishing more for complexity)
- Meaningful only to **compare between models built on the same data set**

2. Evaluation measures for binary classification



Confusion matrix

More on this in Chapter 10: Cost-sensitive learning and decision-making

- The **Confusion (CF) matrix**:
  - **Negative**: **0**
    - Good, no churn, no fraud,…
  - **Positive**: **1**
    - Bad, churn, fraud, …
  - Meaning: matter of convention
- Or, the transposed matrix
  - Matter of convention
  - Elkan (2001):
    - Rows predicted class
    - Columns actual class
- Or, with the rows switched (cf. infra)

CF		Actual class	
		Negative	Positive
Predicted class	Negative	True Negative (TN)	False Negative (FN)
	Positive	False Positive (FP)	True Positive (TP)

CF		Predicted class	
		Negative	Positive
Actual class	Negative	True Negative (TN)	False Positive (FP)
	Positive	False Negative (FN)	True Positive (TP)

Elkan, C. (2001, August). The foundations of cost-sensitive learning. In *International joint conference on artificial intelligence* (Vol. 17, No. 1, pp. 973-978). Lawrence Erlbaum Associates Ltd.

Performance metrics: example

Issues?

Test set:

	Actual	Score = P(y=1|x)
John	Bad	0,84
Emma	Good	0,76
David	Bad	0,48
Sophie	Good	0,44
Bob	Good	0,18

Cut-off = 0,50



	Actual	Score	Predicted
John	Bad	0,84	Bad
Emma	Good	0,76	Bad
David	Bad	0,48	Good
Sophie	Good	0,44	Good
Bob	Good	0,18	Good

		Actual class	
		Negative	**Positive**
Predicted class	**Negative**	**True Negative** **2: Sophie, Bob**	**False Negative** **1: David**
	Positive	**False Positive** **1: Emma**	**True Positive** **1: John**

- PCC = ACC = (TP+TN)/(TP+FP+FN+TN) = 3/5
- ERR = (FP+FN)/(TP+FP+FN+TN) = 2/5
- SENS = TP/(TP+FN) = 1/2
- SPEC = TN/(FP+TN) = 2/3

Performance metrics

CF		Actual class	
		Negative	Positive
Predicted class	Negative	True Negative (TN)	False Negative (FP)
	Positive	False Positive (FN)	True Positive (TN)

- Performance metrics based on the confusion matrix:
  - Correctly classified instances = (TP + TN)
  - Incorrectly classified instances = (FP + FN)

  - Classification accuracy (ACC) = percentage correctly classified (PCC)
    = (TP + TN) / (TP + TN + FP + FN)
  - Classification error (ERR) or misclassification rate:
    = (FP + FN) / (TP + TN + FP + FN)

  - Sensitivity (SENS) = TP / (TP + FN) = Recall
  - Specificity (SPEC) = TN / (FP + TN)

For your information:

Note:
- Here, first row are **positives** and second row are **negatives**
- Matter of convention

		True condition			
	Total population	Condition positive	Condition negative	Prevalence $= \frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) $= \frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$
Predicted condition	Predicted condition positive	**True positive**, Power	**False positive**, Type I error	Positive predictive value (PPV), Precision $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$
	Predicted condition negative	**False negative**, Type II error	**True negative**	False omission rate (FOR) $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR−}}$
		False negative rate (FNR), Miss rate $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR−) $= \frac{\text{FNR}}{\text{TNR}}$	F_1 score $= 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

https://en.wikipedia.org/wiki/Matthews_correlation_coefficient
https://towardsdatascience.com/the-f1-score-bec2bbc38aa6

F1 score

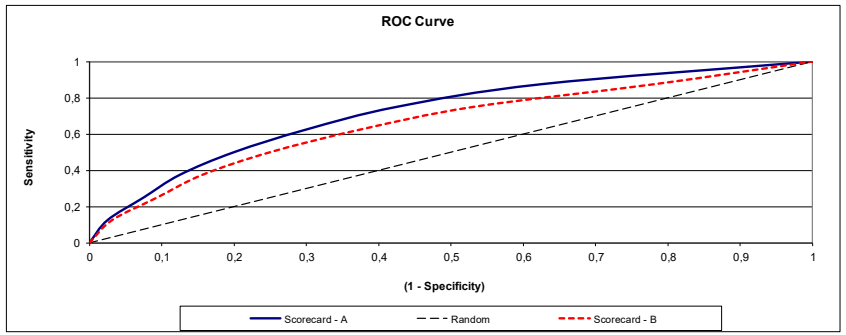
- $F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$
  - **Recall** = Sensitivity = proportion of correctly classified positives
  - **Precision** = proportion of predictive positives that are positive
  - F1 = Harmonic mean of Precision and Recall
  - Single measure that takes into account both performance measures
  - Suitable and often-used **for imbalanced class distribution instead of accuracy**

- F0.5, F2, F3, … Fbeta score:
  - $F_{beta} = (1 + beta^2) \frac{Precision \cdot Recall}{beta \cdot Precision + Recall}$

https://hasty.ai/docs/mp-wiki/metrics/f-beta-score
https://machinelearningmastery.com/fbeta-measure-for-machine-learning/
https://towardsdatascience.com/is-f1-the-appropriate-criterion-to-use-what-about-f2-f3-f-beta-4bd8ef17e285

Receiver Operating Characteristic Curve

Cut-off	sensitivity	specificity	1-specificity
0	1	0	1
0,01			
0,02			
….			
0,99			
1	0	1	0



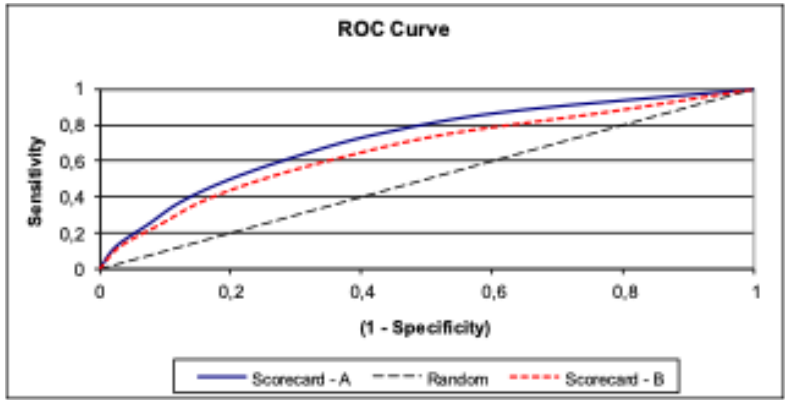
- Make a table with sensitivity and specificity for each possible cutoff
- Receving Operating Characteristic (ROC) curve plots sensitivity versus 1-specificity for each possible cutoff
- 1-Specificity?
  - Sensitivity = detection power, recall
  - 1–Specificity = false alarm rate

- ROC curve of a perfect model?
- ROC curve of a random model?
- Best model?
- What if the curves intersect?

Baseline models: extreme case reasoning – provides insight, feeling

ROC curve and Area under the ROC curve

- AUC = AUROC = Area under the ROC curve
  - Measure of performance
  - Independent of cut-off
  - Interpretation?



An intuitive interpretation of the AUC is that it provides an estimate of the probability that a randomly chosen instance of class 1 (good payer) is correctly ranked higher than a randomly chosen instance of class 0 (bad payer)

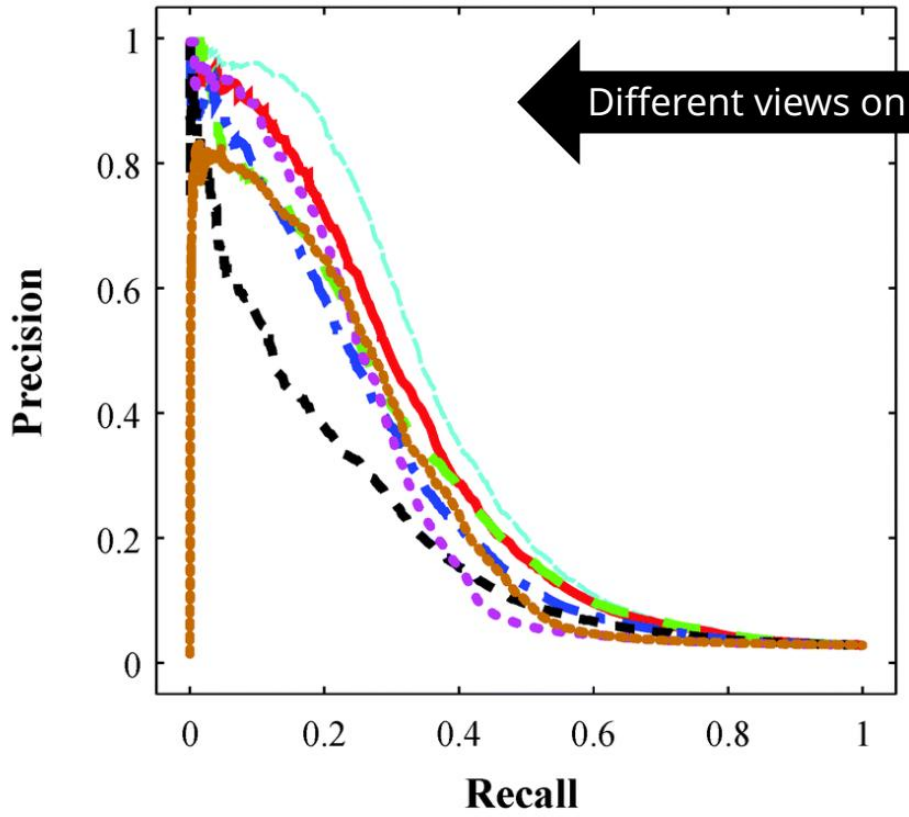
(Hanley and McNeil 1983)

Model performance and validity

Precision-recall curve & AUPRC

- For imbalanced datasets, ignores true negatives
- Baseline: random model has AUPRC = positive class prevalence

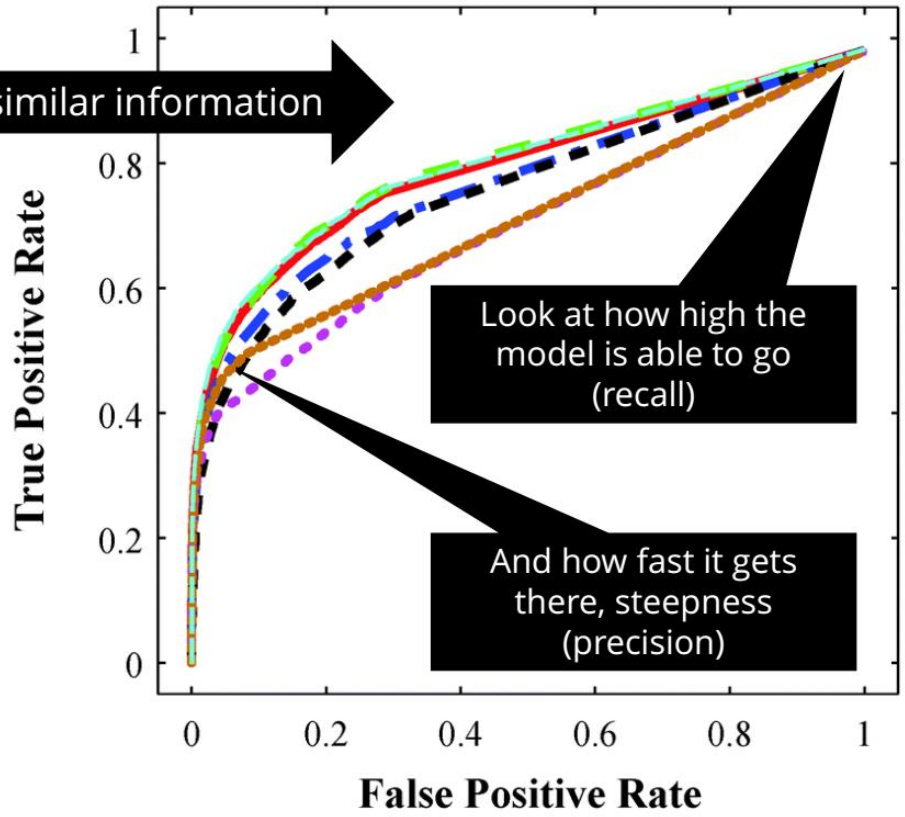
Precision-recall curve



(= sensitivity = true positive rate = detection power)

Different views on similar information

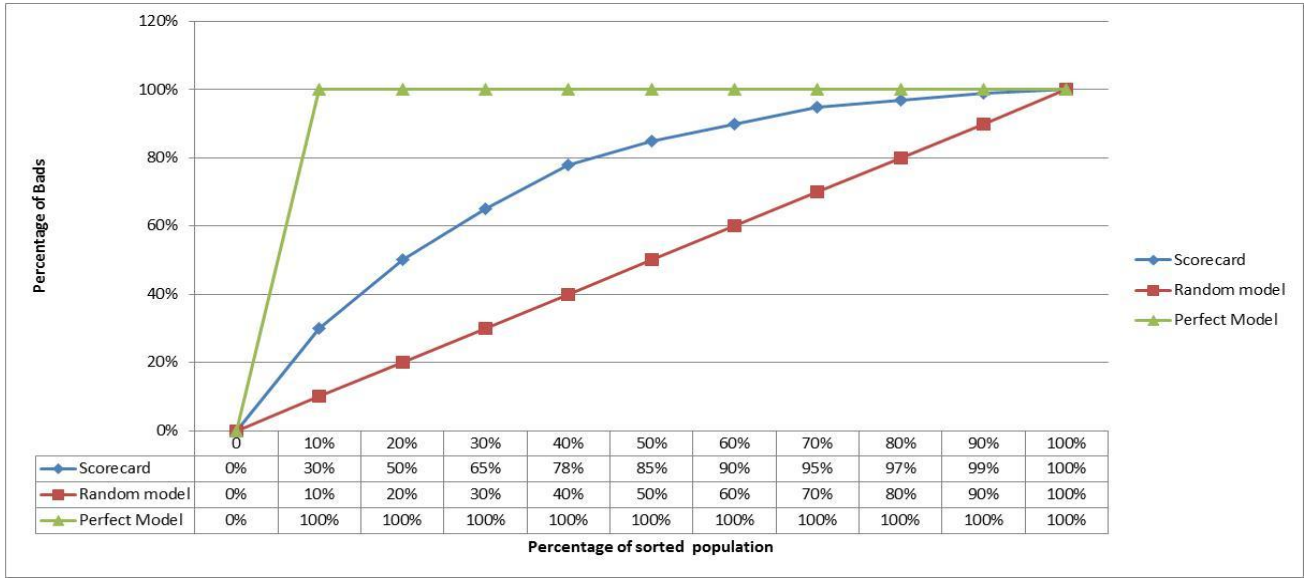
ROC curve



(= 1 - specificity)

Cumulative accuracy profile (CAP) curve

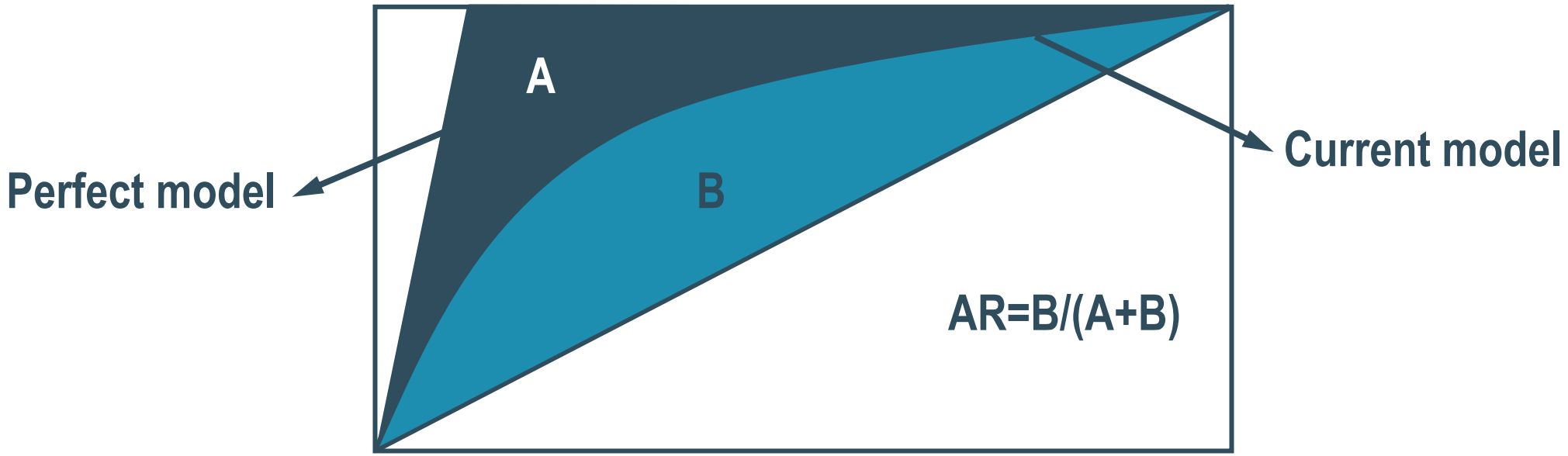
- Sort the population from high probability to be positive to low
- Measure the (cumulative) percentage of positives for each score decile
- For example, the top 30% are most likely positives - model captures 65% of positives
- Also referred to as **Lorenz-**, **power-, or Gini-curve**, or **captured event plot** (SAS)



	0	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Scorecard	0%	30%	50%	65%	78%	85%	90%	95%	97%	99%	100%
Random model	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
Perfect Model	0%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Accuracy ratio or Gini

Area A: dark blue, below the curve of the perfect model and above the curve of the current model
Area B: light blue, above the diagonal below the curve of the current model

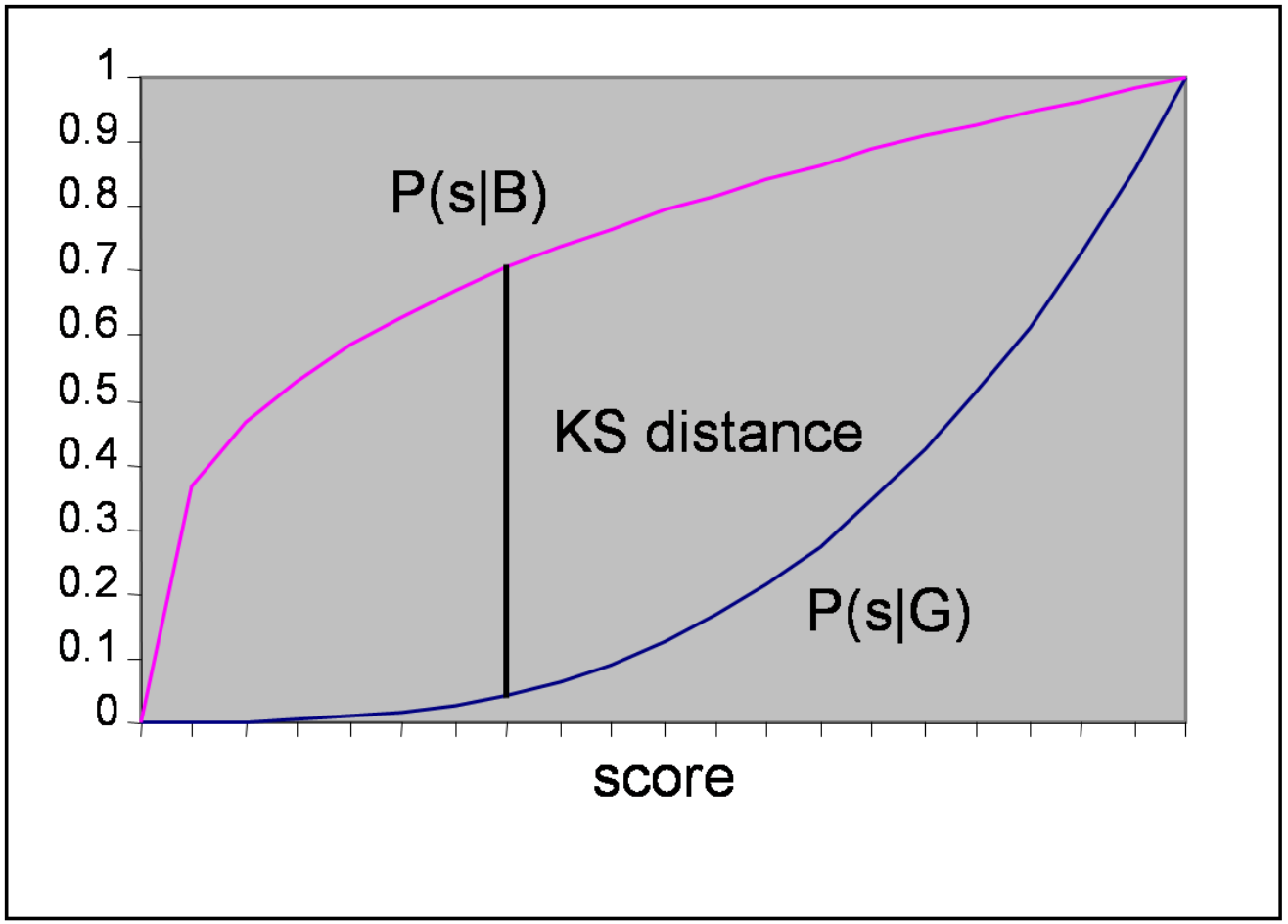


- The **Accuracy Ratio (AR)** or **Gini** is defined as the ratio of area B over the area A + B, i.e.:

$$\frac{\text{(Area below power curve for current model-Area below power curve for random model)}}{\text{(Area below power curve for perfect model-Area below power curve for random model)}}$$

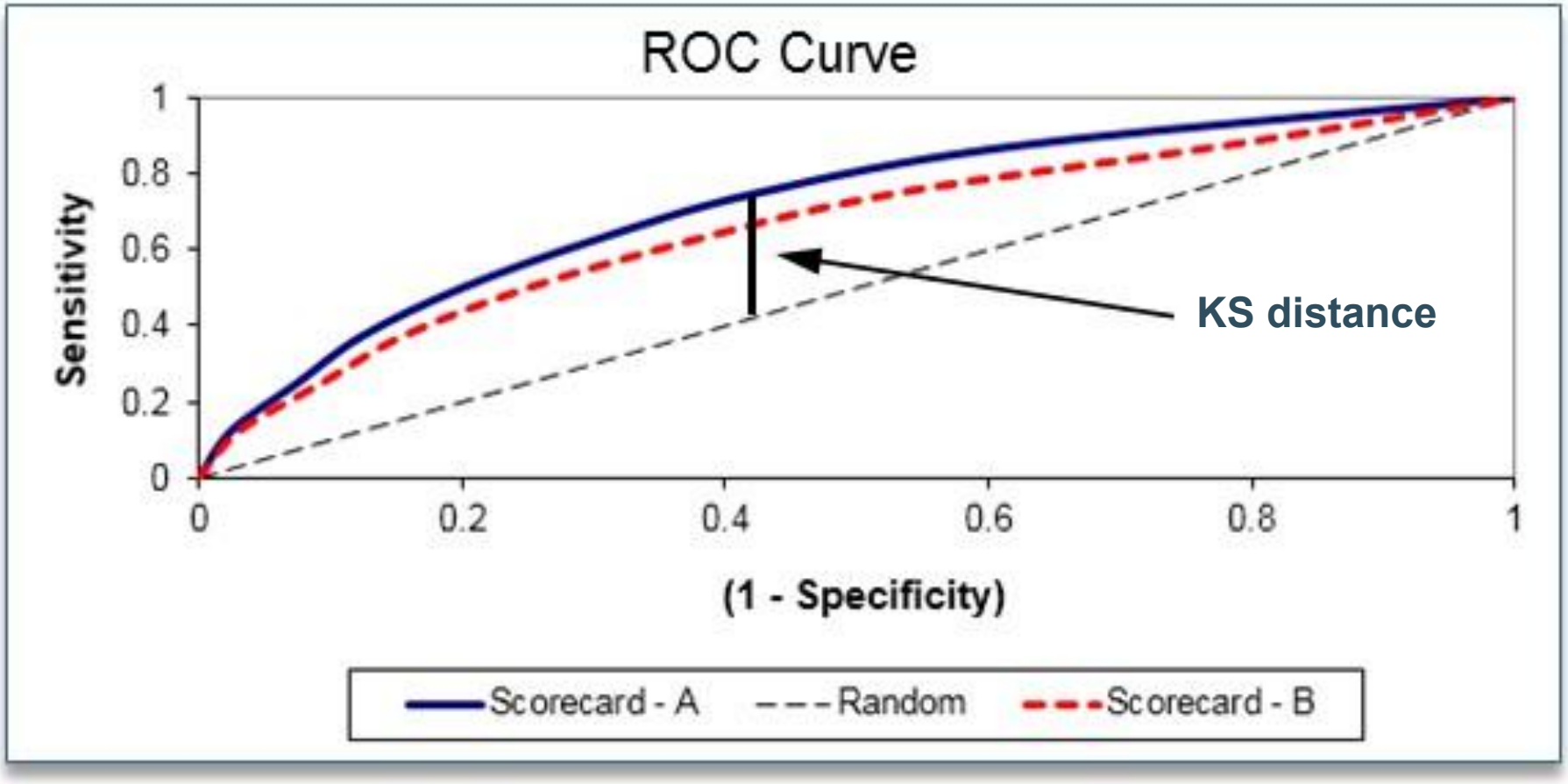
- Perfect model has an AR of 1, Random model has an AR of 0.
- **AR = 2*AUC-1**

Kolmogorov-Smirnov (KS) distance



KS distance and the ROC curve

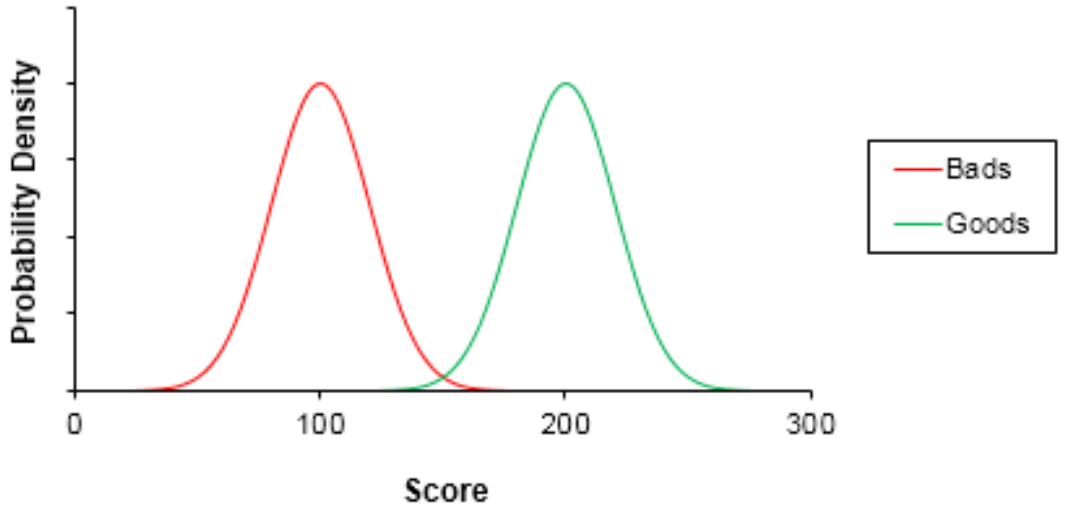
Baseline models?



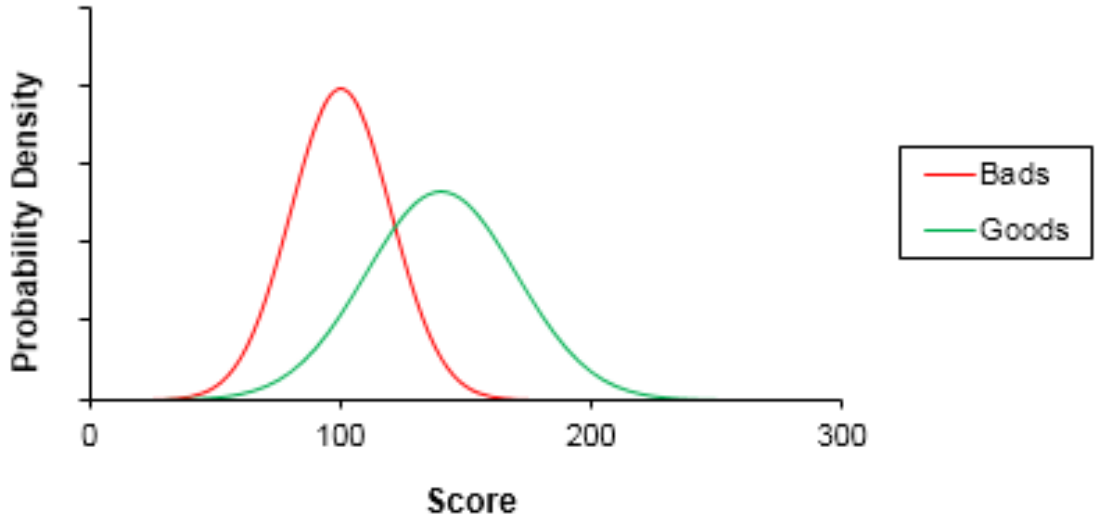
Distance measures

Baseline models?

- Measuring the separation of the distributions of positives and negatives



Good separation



Bad separation

Distance measures

- Euclidean distance does not take the distribution (standard deviation) of the scores into account
- The Mahalanobis distance measures the distance between the two mean scores of the scorecard:

$$M = \left| \frac{\mu_P - \mu_N}{\sigma} \right|$$

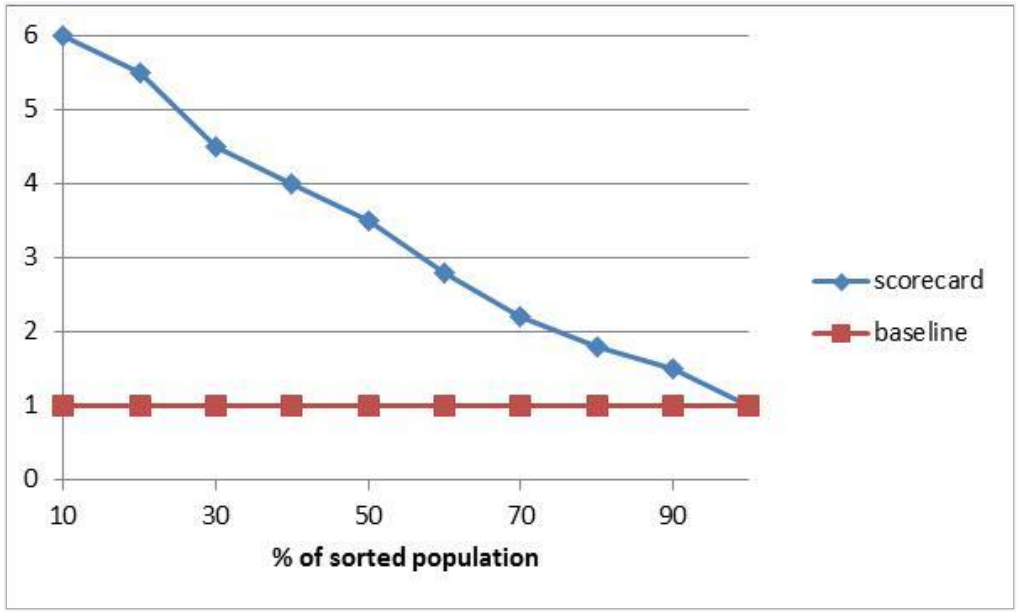
with σ the (pooled) standard deviation of the scores of the goods and the bads from their respective means.

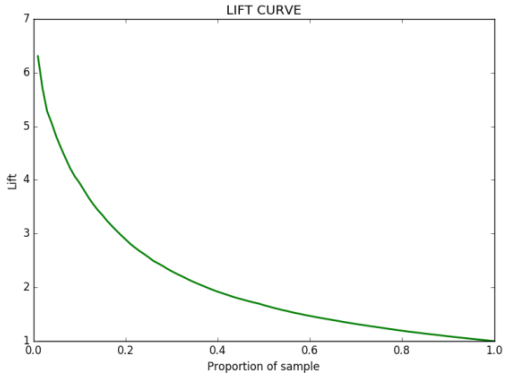
- Closely related is the divergence measure D:

$$D = \frac{(\mu_P - \mu_N)^2}{\frac{1}{2}(\sigma_P^2 + \sigma_N^2)}$$

- Seldomly used

Lift curve





Measures derived from lift curve:

- Top decile lift
- Top five percent lift
- … Lift at cut-off (x%)

- Sort population from low score to high score
- Suppose in top 10% lowest scores, 60% are bads; If in total population, 10% are bad, then lift becomes 60/10=6
- The lift value is thus the **cumulative** percentage of bads per decile, divided by the overall population percentage of bads
- Using no model or random sorting, lift would always be 1!
- Lift can also be expressed in a non-cumulative way

Sample question - medium

Which statement about the ROC curve is **TRUE**?

A. The ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings, and the ROC space is defined by FPR as y axis and TPR as x axis.

B. The best possible prediction method would yield a point with (x,y)-coordinate equal to (1,0) in the ROC space, representing 100\% sensitivity (no false negatives) and 100\% specificity (no false positives). The (1,0) point is also called a perfect classification.

C. The ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings. The true-positive rate is also known as specificity or recall. The false-positive rate is equal to (1 − sensitivity).

D. The ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings. The true-positive rate is also known as sensitivity or recall. The false-positive rate is equal to (1 − specificity).

PollEv.com/wouterverbeke375

Sample question - difficult

Which of the following statements is **FALSE**? Give answer D if all statements are true.

A. The evaluation measures that are used in the single-sample method for evaluating model performance, e.g., the AIC and BIC measures, evaluate the trade-off between the fit of the model and its complexity. They are only meaningful to compare the performance of two models that are built on the same data set.

B. An intuitive interpretation of the AUC performance measure is that it provides an estimate of the probability that a randomly chosen instance of the positive class is correctly ranked higher than a randomly chosen instance of the negative class.

C. For evaluating the performance of a classification model, distance measures such as the divergence measure or Euclidean distance can be used for measuring the separation of the distributions of positives and negatives, taking into account both the difference between the average value of the probability estimates and the variability around the mean of the estimates.

D. All the above statements are true.

PollEv.com/wouterverbeke375

3. Evaluation measures for multiclass classification

Evaluation measures for multiclass classification

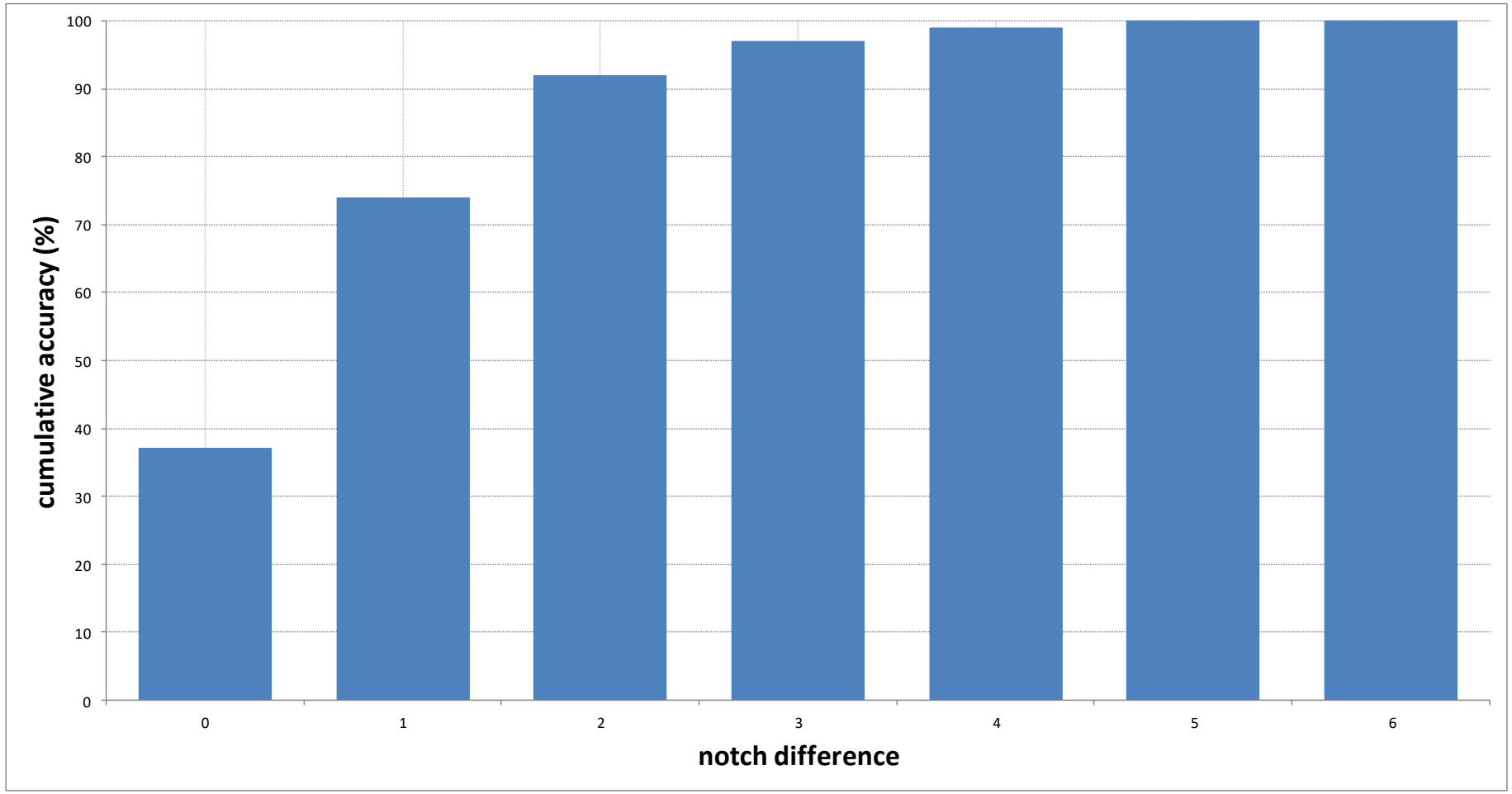
- For example, rating problems
- Classes are A+, A, A-, B+, B, B-, ...
- Confusion matrix
  - No specificity, sensitivity
- Notch difference graph
- Area under the ROC curve

Confusion matrix

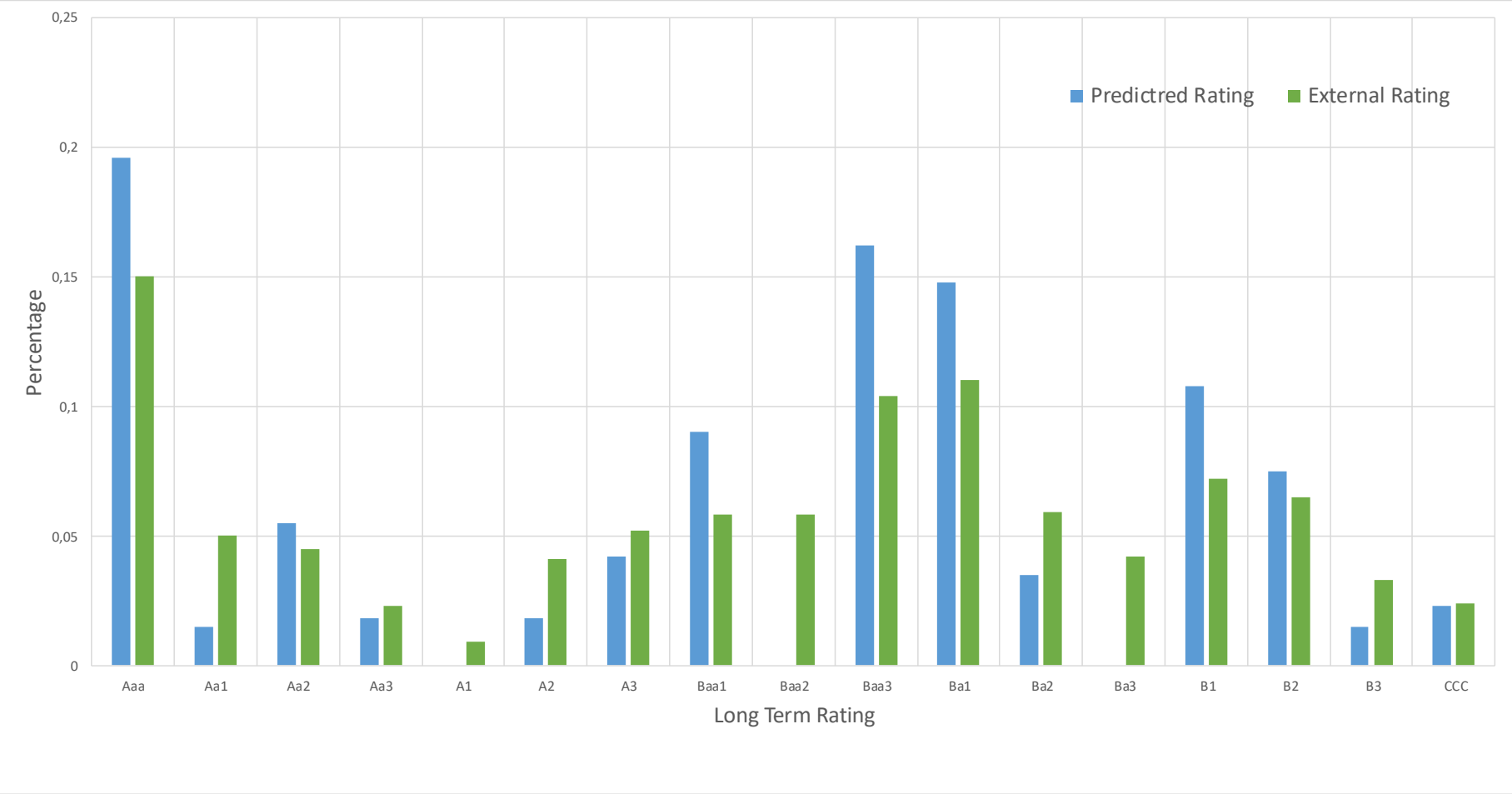
TRUE \ PREDICTED	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E+	E	E-
A+	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	0	14	0	4	0	0	0	0	0	0	0	0	0	0	0
A-	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0
B+	0	7	0	50	0	0	5	5	0	1	0	0	0	0	0
B	0	2	2	54	0	12	58	17	0	5	1	0	1	0	0
B-	0	0	0	6	0	6	10	6	0	1	0	0	0	0	0
C+	0	0	0	13	0	3	99	34	0	11	1	0	0	0	0
C	0	1	0	5	0	1	37	151	1	12	4	0	0	0	0
C-	0	0	0	2	0	0	3	8	3	4	0	0	0	0	0
D+	0	0	0	1	0	0	7	20	2	131	24	0	3	1	0
D	0	0	0	1	0	0	2	9	1	32	122	0	13	7	0
D-	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0
E+	0	0	0	0	0	0	0	4	0	12	26	0	37	5	0
E	0	0	0	0	0	0	0	4	0	4	9	0	5	48	0
E-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- Not all errors have equal impact
- The farther away from the diagonal, the bigger the impact of the error!

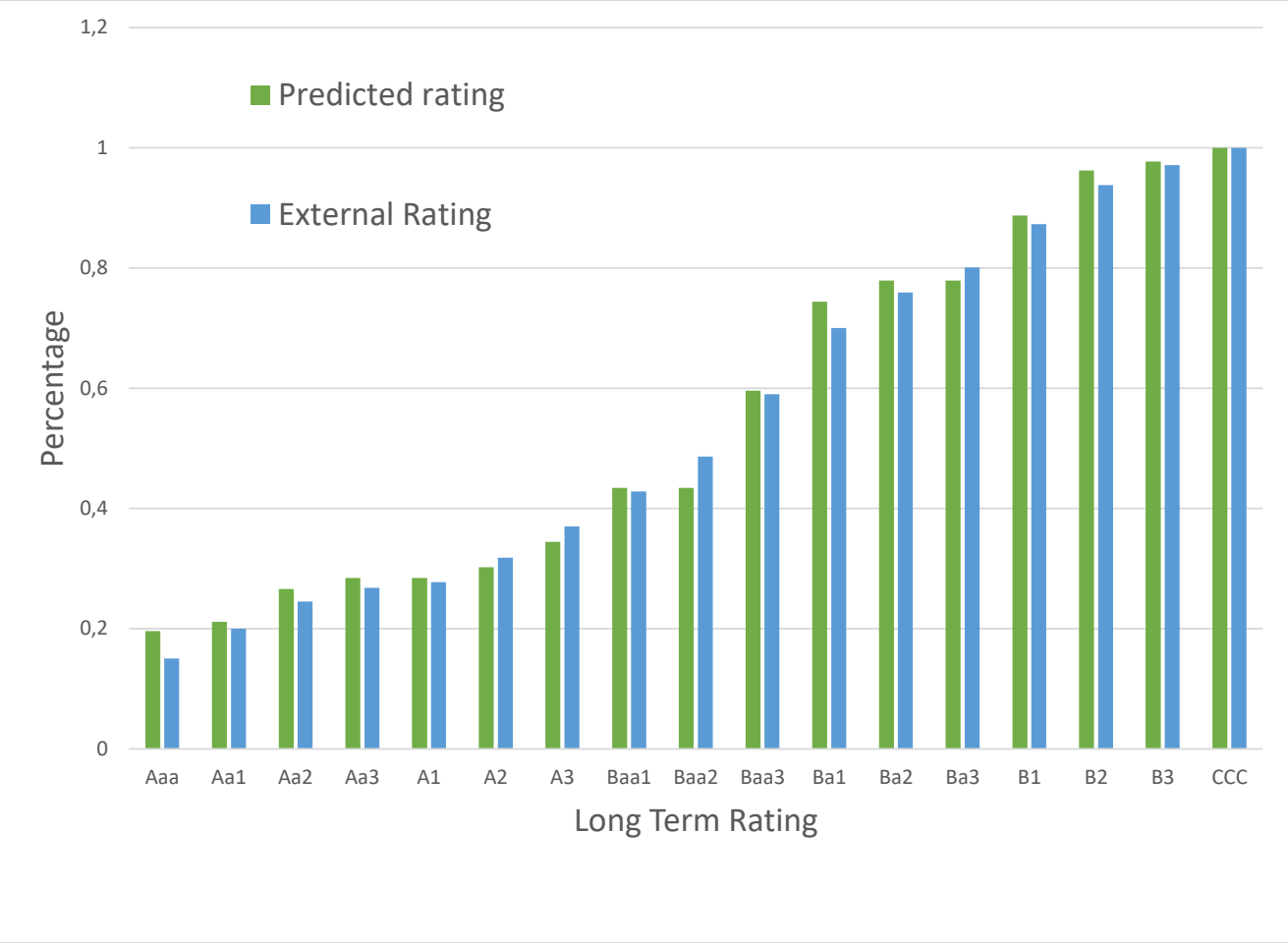
Notch difference graph



Distributions of classes



Distributions of classes



AUC for multiclass

- Method 1:
  - Produce n different ROC graphs, one for each class (other classes merged into one) and calculate AUC.
  - Calculate overall AUC as weighted average of individual AUCs (use prior probability of class as weights).
- Method 2:
  - Calculate $\binom{n}{2}$ AUCs, one for each possible class comparison and take the average.
- References
  - Fawcett, T., *ROC Graphs: Notes and Practical Considerations for Researchers*, HP Labs Tech Report HPL-2003–4.
  - Hand, D. and Till, R.J., "A simple generalization of the area under the ROC curve to multiple class classification problems," *Machine Learning*, 45(2), pp. 171–186, 2001.

4. Evaluation measures for regression



Evaluation measures for regression

- R-squared:
  - Adjusted R-squared (cf. AIC, BIC)

$$Adj.R^2 = 1 - \left[(1 - R^2) \cdot \frac{n-1}{n-k-1}\right]$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- Mean Squared Error:
  - RMSE

$$MSE = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n}$$

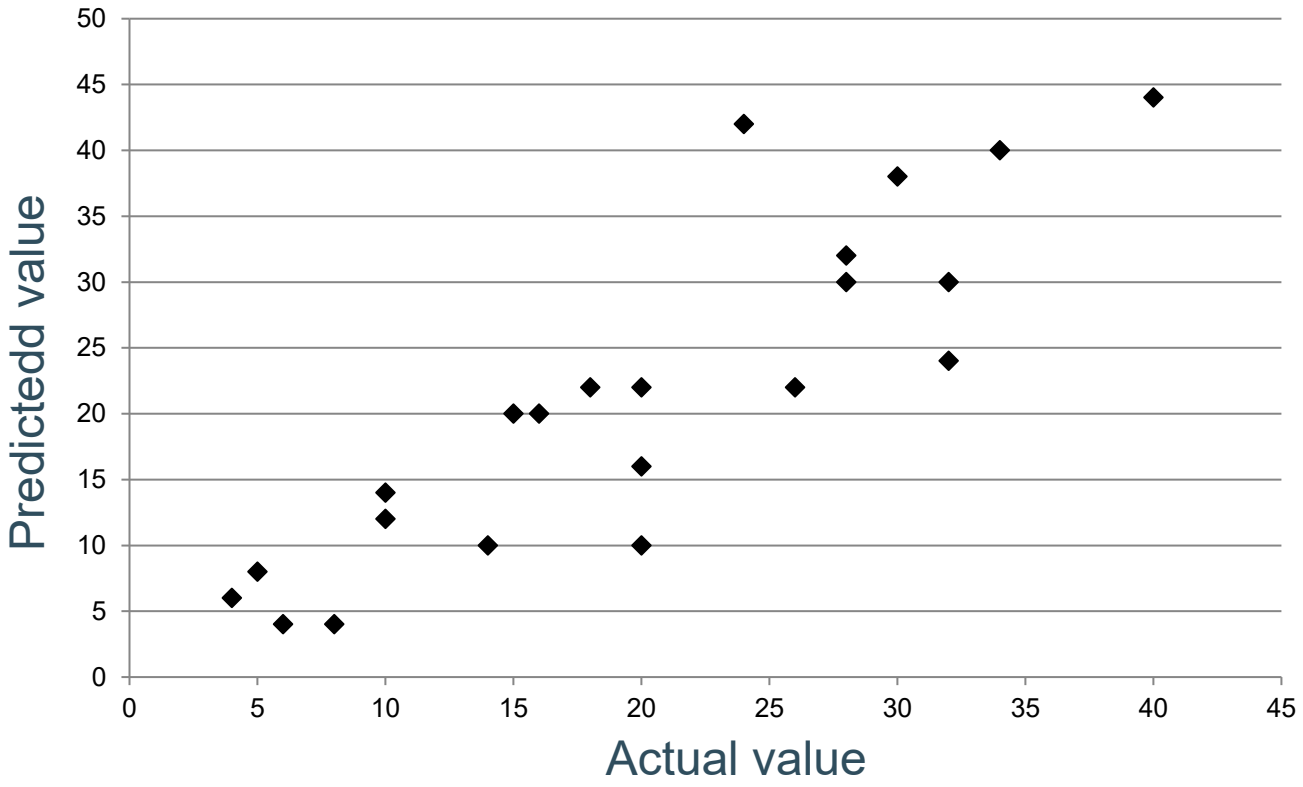
- Mean Absolute Deviation:

$$MAD = \frac{\sum_{i=1}^{n}|y_i - \hat{y}_i|}{n}$$

- With y_i the actual outcome, $\hat{y}_i$ the estimated outcome, and $\bar{y}$ the average outcome

https://www.statisticshowto.com/probability-and-statistics/statistics-definitions/adjusted-r2/

Evaluation measures for regression



$$\rho_P = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

Q: Could we use correlation for evaluating a binary classification model?

Evaluation measures for regression

- Re-using classification measures?

Data	
Y	f(X)
12%	37%
82%	75%
54%	43%
42%	59%
5%	15%
98%	92%
…	…

Threshold = 50%	
Y	f(X)
0	0,37
1	0,75
1	0,43
0	0,59
0	0,15
1	0,92
…	…
AUC =	**0,86**

Threshold = 25%	
Y	f(X)
0	0,37
1	0,75
1	0,43
1	0,59
0	0,15
1	0,92
…	…
AUC =	**0,72**

Threshold = 75%	
Y	f(X)
0	0,37
1	0,75
0	0,43
0	0,59
0	0,15
1	0,92
…	…
AUC =	**0,94**

Evaluation measures for regression

- Re-using classification measures?
  - Turn continuous target variable in a binary target variable:
    - Adopt cut-off, e.g., equal to the median value
    - Values above the median are positive class observations, values below the median are negative class observations
  - Predicted value is score: ROC curve and AUC can be used, or other measures
  - Indicates whether the model discriminates well between high and low values
  - High and low defined by the adopted cut-off, e.g., the median
  - Alternative cut-offs:
    - Q1: detection power for very low values
    - Q3: detection power for very high values
    - …

KU LEUVEN

Chapter 5. Postprocessing

Data Science for Business

D0I74A

Prof. dr. Wouter Verbeke

Prof. dr. Bart Baesens



Course schedule

BLOCK1			
1	21/09/2026	Chapter 0. Course Outline Chapter 1. Data Science for Business	
1	22/09/2026	Chapter 1. Data Science for Business (ctd.)	
		Hands-on #1: Building a first pipeline	**Homework 1**
2	28/09/2026	Chapter 1. Data Science for Business (ctd.) Chapter 2. Exploratory Data Analysis & Preprocessing	
2	29/09/2026	Chapter 2. Exploratory Data Analysis & Preprocessing (ctd.)	
		Hands-on #2: EDA & Preprocessing	**Homework 2**
3	05/10/2026	Chapter 2. Exploratory Data Analysis & Preprocessing (ctd.)	
3	06/10/2026	Chapter 3. Predictive Analytics - Decision Trees	
4	12/10/2026	Chapter 4. Predictive Analytics - Nearest Neighbors & Ensembles	
4	13/10/2026	Chapter 5. Postprocessing	
5	19/10/2026	Chapter 5. Postprocessing	
		Assignment 1	**Report 1: 10/11**

Course schedule

BLOCK2			
6	26/10/2026	Chapter 6. Predictive Analytics - Artificial Neural Networks	
6	27/10/2026	Chapter 7. Predictive Analytics - Forecasting	
7	02/11/2026	Chapter 8. Descriptive Analytics	
7	03/11/2026	Chapter 9. Prescriptive Analytics - Causal Machine Learning	
8	09/11/2026	Chapter 9. Prescriptive Analytics - Causal Machine Learning	
8	10/11/2026	Chapter 10. Prescriptive Analytics - Profit-driven Analytics	**Report 1 deadline**
9	16/11/2026	Chapter 11. Prescriptive Analytics - Multi-armed Bandits & Reinforcement Learning	
9	17/11/2026	Guest Lecture #1 Toon Vanderschueren, Ph.D., Causal ML @Telenet	**Presence expected**
	17/11/2026	**Assignment 2**	**Report 2 > 14/12**
BLOCK3			
10	23/11/2026	Chapter 12. Natural Language Processing	
10	24/11/2026	Guest Lecture #2 Dr. Véronique Van Vlasselaer, Ph.D., Business Analytics @SAS	**Presence expected**
11	30/11/2026	Chapter 12. Natural Language Processing	
11	01/12/2026	Chapter 13. Large Language Models & AI Agents	
12	07/12/2026	Chapter 13. Large Language Models & AI Agents	
12	08/12/2026	BACK-UP	
13	14/12/2026	Feedback Assignment 2 Sample Exam Questions	**Report 2 deadline**
13	15/12/2026	Guest Lecture #3 To be confirmed	**Presence expected**

Outline

1. Evaluation methodology
2. Evaluation measures for binary classification
3. Evaluation measures for multiclass classification
4. Evaluation measures for regression
5. Evaluation beyond predictive accuracy
6. Case: Credit risk modeling
   - Scorecards
   - Ratings and calibration
   - Backtesting and benchmarking
7. XAI

5. Evaluation beyond predictive accuracy

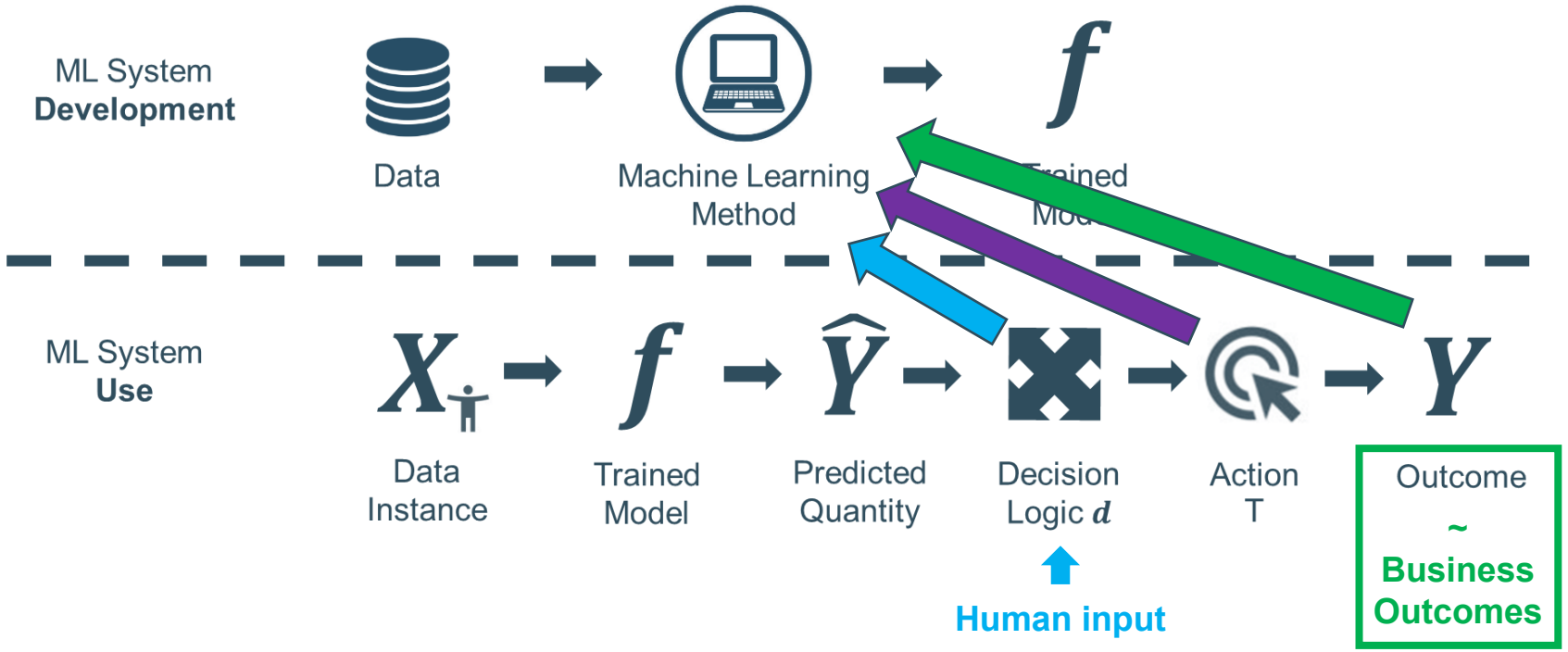


Machine learning for business decision-making

A prediction is not a decision!

→ **Decision logic d** with $T = d(\hat{Y}, \dots)$ (~policy, optimization, rules, …)

- Determines **how predictions result in actions or decisions**
- Taking into account **business objectives, constraints and requirements**
- **Benefits in integrating with ML System development? How to?**



Decision-centric learning
Predict-then-optimize
Predict-and-optimize
…

Causal machine learning
Reinforcement learning
Prescriptive analytics
…

Cost-sensitive learning
Profit-driven analytics
…

Credits for figure to prof. Foster Provost

Evaluation beyond predictive accuracy

- Evaluation criterions for assessing quality of an analytical model
  - Statistical evaluation
    - Accuracy, …
    - Statistical significance
  - Interpretability, explainability, justifiability
    - Scorecard format, variables and relations, …

    White-box vs. black-box
  - Economic cost
    - Cf. Chapter 10: Prescriptive analytics: cost-sensitive evaluation and decision-making
    - External data, ROA, CO^2 emissions, energy usage, computation time, …
  - Operational efficiency
    - Feasability
  - Regulation

    GDPR, Basel, Solvency, …
  - ...
- **Does it solve the business problem?**
  - Cf. Step 1 in the analytics process model!

An *appropriate* evaluation is case-specific!

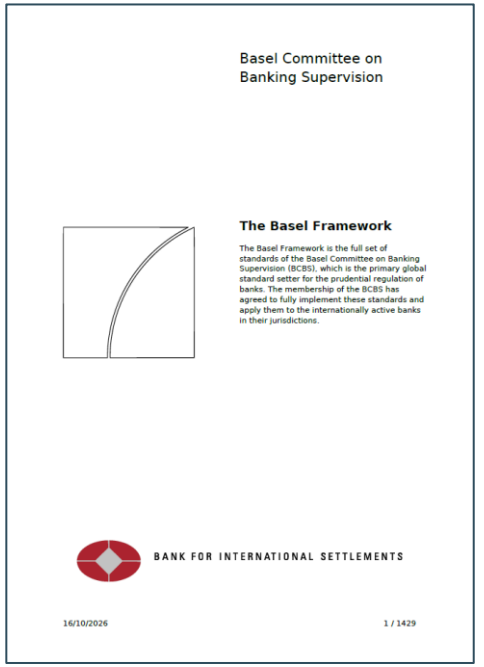
6. Case: Credit risk modeling

Scorecards

Ratings & Calibration

Backtesting & benchmarking

Case: Credit risk modeling



- Basel regulation (or Basel Capital accord)
  - Imposing *standard approach* to calculate regulatory capital requirements
  - **Standardized approach**:
    - Ratings from rating agencies (S&P, Moody's, …)
    - Risk weights in accord to calculate risk weighted assets (RWA)
    - Capital requirement = 8%.RWA
  - **Internal ratings based (IRB) approach**:
    - Financial institutions develops a rating system (credit risk models) to estimate for *every* exposure:
      - Probability of default (PD)
      - Loss given default (LGD)
      - Exposure at default (EAD)
    - Allows to calculate:
      - **Expected Loss (EL) = PD x LGD x EAD**
      - Capital requirements = unexpected losses 99.9%), calculated using the Basel **risk weight function**
- IFRS9 regulation: accounting standards, forward looking, requiring similar estimates

https://www.bis.org/committees/bcbs/history
https://www.bis.org/committees/bcbs/basel-framework

Case: Credit risk modeling

For every exposure, we need estimates for PD, LGD, EAD!

- **Risk weight function** (for retail exposures):

$$K = \boldsymbol{LGD}.\left(N\left(\sqrt{\frac{1}{(1-\rho)}} N^{-1}(\boldsymbol{PD}) + \sqrt{\left(\frac{\rho}{1-\rho}\right)} N^{-1}(0.999) \right) - \boldsymbol{PD} \right)$$

  - N(): Cumulative standard normal distribution
  - $N^{-1}()$: Inverse cumulative standard normal distribution
  - ρ: Asset correlation

    Residential mortgage exposures: $\rho = 0.15$

    Qualifying revolving exposures: $\rho = 0.04$

    Other retail exposures: $\rho = 0.03\left(\frac{1-e^{-35PD}}{1-e^{-35}}\right) + 0.16\left(1 - \frac{1-e^{-35PD}}{1-e^{-35}}\right)$

- **Regulatory capital = K.EAD**

Case: Credit risk modeling

- **IRB model architecture**

- Why **scorecards**?
  - Implementation software
  - Ease of interpretation
    - No intercept, logarithmic transformations, …
  - Continuity with existing scorecards
  - Legal requirements
    - Explain decision

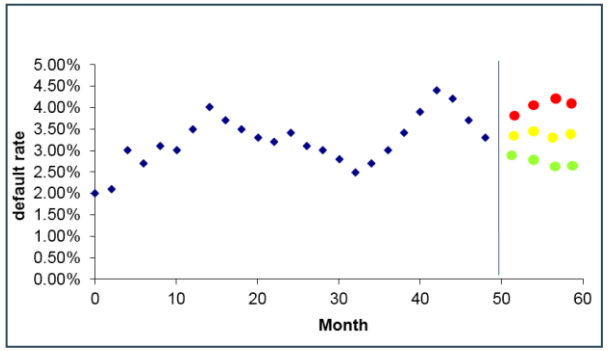
Level 2: Rating system
→ Risk **ratings** definition
→ PD/LGD/EAD **calibration**

↕

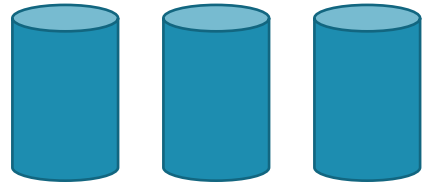
Level 1: Model
→ E.g., Logistic regression
→ **Scorecard**

↕

Level 0: Data
→ Internal Data
→ External Data
→ Expert Judgment



Characteristic Name	Attribute	Points
AGE 1	Up to 26	100
AGE 2	26 - 35	120
AGE 3	35 - 37	185
AGE 4	37+	225
KNOWN CUSTOMER	NO	90
KNOWN CUSTOMER	YES	180
SALARY 1	Up to 500	120
SALARY 2	501-1000	140
SALARY 3	1001-1500	160
SALARY 4	1501-2000	200
SALARY 5	2000+	240



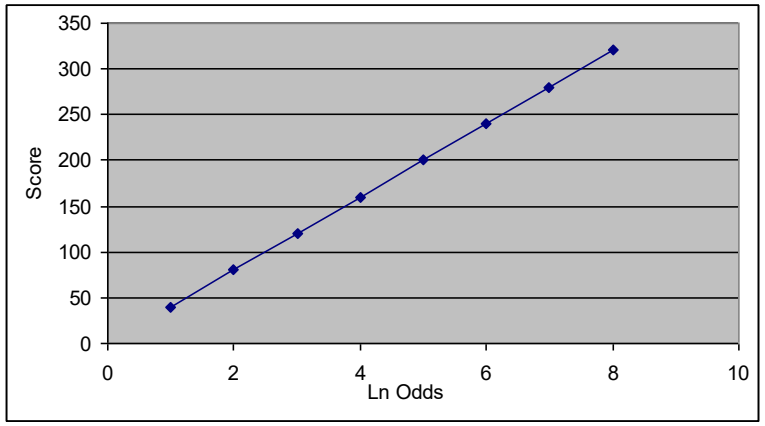
Scorecards

- Which decisions to make in creating a scorecard?

- Characteristics to include
  - Variable selection
- Levels
  - Binning, categorization
- Points
  - Logistic regression + WOE!
  - **HOW?**

Characteristic	Level	Points
AGE 1	Up to 26	100
AGE 2	26 - 35	120
AGE 3	35 - 37	185
AGE 4	37+	225
KNOWN CUSTOMER	NO	90
KNOWN CUSTOMER	YES	180
SALARY 1	Up to 500	120
SALARY 2	501-1000	140
SALARY 3	1001-1500	160
SALARY 4	1501-2000	200
SALARY 5	2000+	240

Scorecards

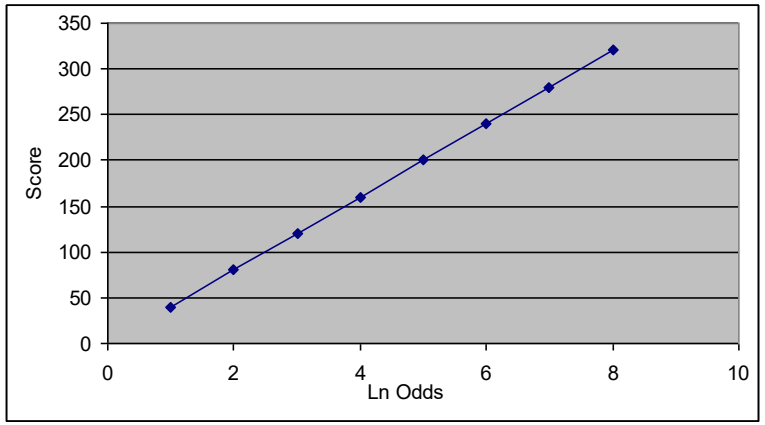


- **Step 1: Determine the scorecard scale**
  - I.e., how odds translate in scores
  - I.e., the linear transformation of ln odds in score ($y = a + b \cdot x$)

$$\text{Score} = \text{offset} + \text{factor} \cdot \ln(Odds)$$

  - To establish the offset and factor (intercept and slope), i.e., the scorecard scale, decide on:
    - Good/Bad odds at a score
    - Points to double the odds
  - Example: to align with the current scorecard in use in a bank:
    - Good/Bad odds should be equal to 50/1 at score 600
    - Good/Bad odds are to double every 20 points

Scorecards



- **Step 2: Calculate the offset and factor**
  - Example:
    - Good/Bad odds should be equal to 50/1 at score 600
    - Good/Bad odds are to double every 20 points
    - Results in two equations:
      - $600 = \text{offset} + \text{factor} \cdot \ln(50)$
      - $620 = \text{offset} + \text{factor} \cdot \ln(100)$
    - This allows to back-out the offset and factor (two equations with two unknowns):
      - $\text{factor} = \frac{20}{\ln(50)}$
      - $\text{offset} = 600 - \text{factor} \cdot \ln(50)$

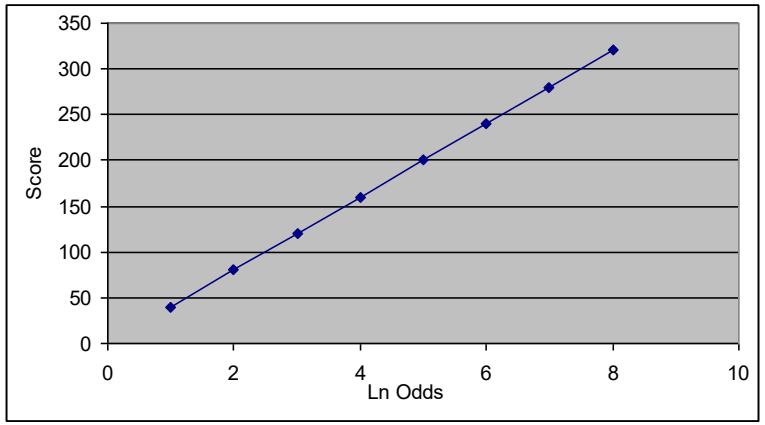
Scorecards

- **Step 3: Plug in the logistic regression model**
  - Logistic regression model:

$$\ln(Odds) = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \cdots + \beta_n \cdot x_n$$
$$\ln(Odds) = \beta_0 + \sum_{i=1}^{n}(\beta_i \cdot x_i)$$
$$\ln(Odds) = \beta_0 + \sum_{i=1}^{n}(\beta_i \cdot woe_i)$$

  - Plug model into the linear transformation:

$$\text{Score} = \text{offset} + \text{factor} \cdot \ln(Odds)$$
$$\text{Score} = \text{offset} + \text{factor} \cdot (\beta_0 + \sum_{i=1}^{n}(\beta_i \cdot woe_i))$$
$$\text{Score} = \text{offset} + \text{factor} \cdot \left(\sum_{i=1}^{n}\left(\frac{\beta_0}{n} + \beta_i \cdot woe_i\right)\right)$$
$$\text{Score} = \sum_{i=1}^{n}\left(\frac{\text{offset}}{n} + \text{factor} \cdot \left(\frac{\beta_0}{n} + \beta_i \cdot woe_i\right)\right)$$



Characteristic	Level	Points
AGE 1	Up to 26	100
AGE 2	26 - 35	120
AGE 3	35 - 37	185
AGE 4	37+	225
KNOWN CUSTOMER	NO	90
KNOWN CUSTOMER	YES	180
SALARY 1	Up to 500	120
SALARY 2	501-1000	140
SALARY 3	1001-1500	160
SALARY 4	1501-2000	200
SALARY 5	2000+	240

Scorecards

Characteristic	Level	Points
AGE 1	Up to 26	100
AGE 2	26 - 35	120
AGE 3	35 - 37	185
AGE 4	37+	225
KNOWN CUSTOMER	NO	90
KNOWN CUSTOMER	YES	180
SALARY 1	Up to 500	120
SALARY 2	501-1000	140
SALARY 3	1001-1500	160
SALARY 4	1501-2000	200
SALARY 5	2000+	240

- **Step 4: Calculate points for characteristic levels**
  - The total score of an applicant is obtained by calculating:

$$\text{Score} = \sum_{i=1}^{n} \left(\frac{\text{offset}}{n} + \text{factor} \cdot \left(\frac{\beta_0}{n} + \beta_i \cdot woe_i \right) \right)$$

  - Then the points of a level of a characteristic can be calculated by entering the values of the offset and factor, and the WOE-value of that level:

$$points = \frac{\text{offset}}{n} + \text{factor} \cdot \left(\frac{\beta_0}{n} + \beta_i \cdot woe_i \right)$$

Ratings and calibration

- Level 2 of credit risk model architecture
  - **Step 1**: Define ratings
  - **Step 2**: Calibrate risk estimates for each rating

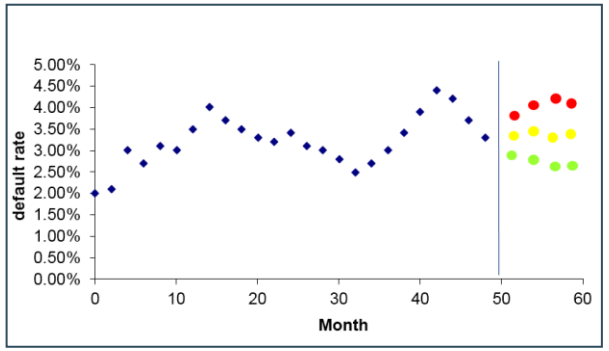
Level 2: Rating system
→ Risk ratings definition
→ PD/LGD/EAD **calibration**

↕

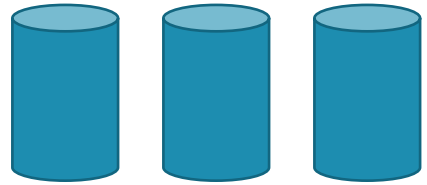
Level 1: Model
→ E.g., Logistic regression
→ **Scorecard**

↕

Level 0: Data
→ Internal Data
→ External Data
→ Expert Judgment



Characteristic Name	Attribute	Points
AGE 1	Up to 26	100
AGE 2	26 - 35	120
AGE 3	35 - 37	185
AGE 4	37+	225
KNOWN CUSTOMER	NO	90
KNOWN CUSTOMER	YES	180
SALARY 1	Up to 500	120
SALARY 2	501-1000	140
SALARY 3	1001-1500	160
SALARY 4	1501-2000	200
SALARY 5	2000+	240



Ratings and calibration

- **Step 1**: Define ratings
  - Rating is a homogeneous pool of obligors that are **similar in terms of default risk**
  - Ratings are defined because **scores are considered too fine granular or too detailed**
  - **Problem statement**: categorize (i.e., bin or segment) “score” obtained from model
    - E.g., from logistic regression model: $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \ldots + \beta_n x_n$

- Approaches
  - Map onto rating agency scale
  - Exponential risk evolution
  - …

Ratings and calibration

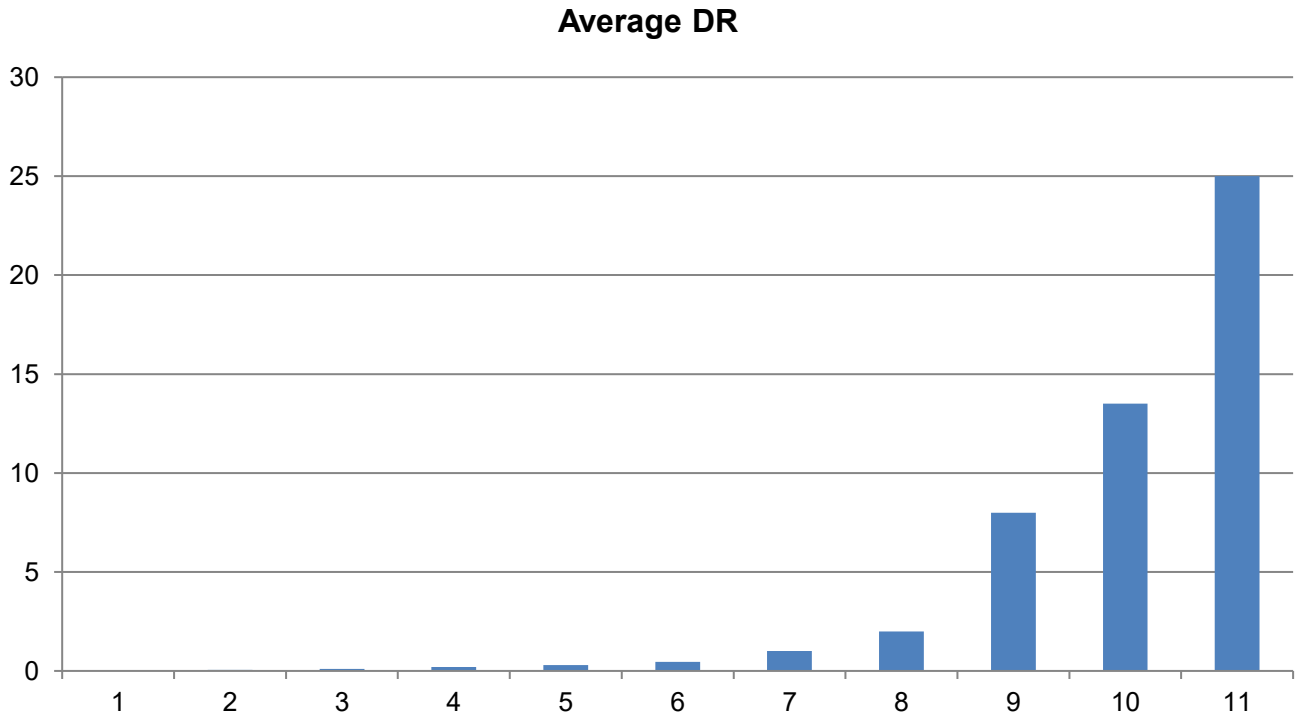
- **Map onto rating agency scale**: define internal ratings so as to mimick as closely as possible 1-year default rates reported by rating agencies

LTRating	Moody's	Fitch	S&P	Average
AAA	0.00%	0.00%	0.00%	0.00%
AA+	0.00%	0.00%	0.00%	0.00%
AA	0.00%	0.00%	0.00%	0.00%
AA–	0.02%	0.00%	0.02%	0.01%
A+	0.00%	0.00%	0.05%	0.02%
A	0.03%	0.00%	0.04%	0.02%
A–	0.04%	0.12%	0.04%	0.07%
BBB+	0.17%	0.29%	0.21%	0.22%
BBB	0.16%	0.13%	0.30%	0.20%
BBB–	0.34%	0.56%	0.39%	0.43%
BB+	0.75%	0.91%	0.65%	0.77%
BB	0.78%	1.80%	0.96%	1.18%
BB–	2.07%	1.94%	1.80%	1.94%
B+	3.22%	1.82%	3.21%	2.75%
B	5.46%	1.78%	8.87%	5.37%
B–	10.46%	1.69%	12.99%	8.38%
CCC	20.98%	26.07%	31.08%	26.04%
Inv.Gr.	0.08%	0.11%	0.11%	
Spec.Gr.	5.15%	3.27%	4.65%	
All	1.74%	0.65%	1.61%	

	Default rate	
LTRating	1-year	5-years
Aaa	0.00%	0.09%
Aa	0.01%	0.20%
A	0.02%	0.56%
Baa	0.21%	2.25%
Ba	1.31%	11.85%
B	5.69%	29.73%
Caa	20.98%	57.01%
Inv.Gr.	0.08%	0.93%
Spec.Gr.	5.15%	23.49%
All	1.74%	7.73%

Ratings and calibration

- **Exponential risk evolution**: define ratings such that default rates double per rating decrease



Rating philosophy

"What is rating system supposed to do?"

- "**Rating philosophy** is the term used to describe the **assignment horizon** of a borrower rating system" (FSA, CP 05/03, par. 7.66)
  - Related to how quickly obligors are expected to migrate from one rating to another in response to economic cycles
- Changes in economic conditions will be reflected through (EBA/GL/2017/16):
  - Rating migration
  - Change in yearly default rates
- Needs to be clearly articulated in **bank's rating policy**

Rating philosophy

- **Point-in-Time (PIT) Ratings**
  - Take into account obligor specific, **cyclical**, and non-cyclical information
  - Rating changes rapidly with macroeconomic situation (lots of rating mobility!)
  - PD is best estimate of obligor's default during next 12 months
  - **IFRS 9**
- **Through-the-Cycle (TTC) Ratings**
  - Only take into account **non-cyclical information**
  - Rating robust with respect to macro-economic situation (not much rating mobility)
  - PD is best estimate of obligor's default risk 'on average' across a credit cycle
  - **Basel!**
- **Many hybrids exist**
  - PIT and TTC represent two ends of continuum!

Rating philosophy

Point-in-Time Ratings	Through-the-Cycle Ratings
Migration to lower rating when entering downturn	No migration when entering downturn
Actual default rate in each rating remains unchanged, and rating level PD remains unchanged	Actual default rate in each rating increases, and rating level PD remains unchanged
Due to downgrade, capital requirements increase	No downgrade, so capital requirements remain theoretically unchanged

PD calibration

- **Assign PD to each rating** using **long-term historical data**

- Calibration deals with 3 sources of variability
  - Sampling variability
  - External **economic conditions**
  - Internal effects (e.g., strategy change)

- Calibrated PD = best estimate PD + conservative add-on

- Basel requires availability of at least 5 years of data for calibration

PD calibration

- Approaches:
  - Average (?)
  - Maximum (?)
  - Upper percentile (which?)
  - Use binomial confidence interval to account for sample variability:

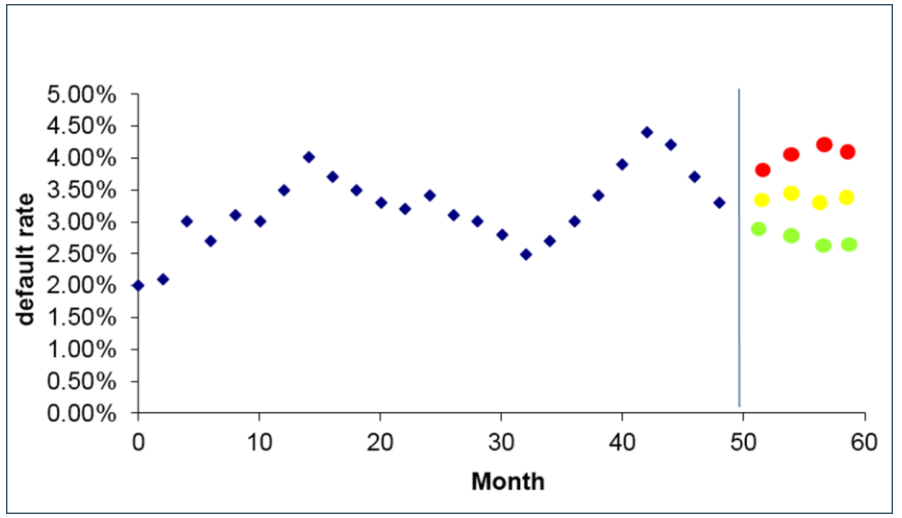
$$\widehat{DR} \pm N^{-1}(\alpha)\sqrt{\frac{\widehat{DR}(1-\widehat{DR})}{N_{Obs}}}$$

  - Using time series model (cf. next slide)

PD calibration

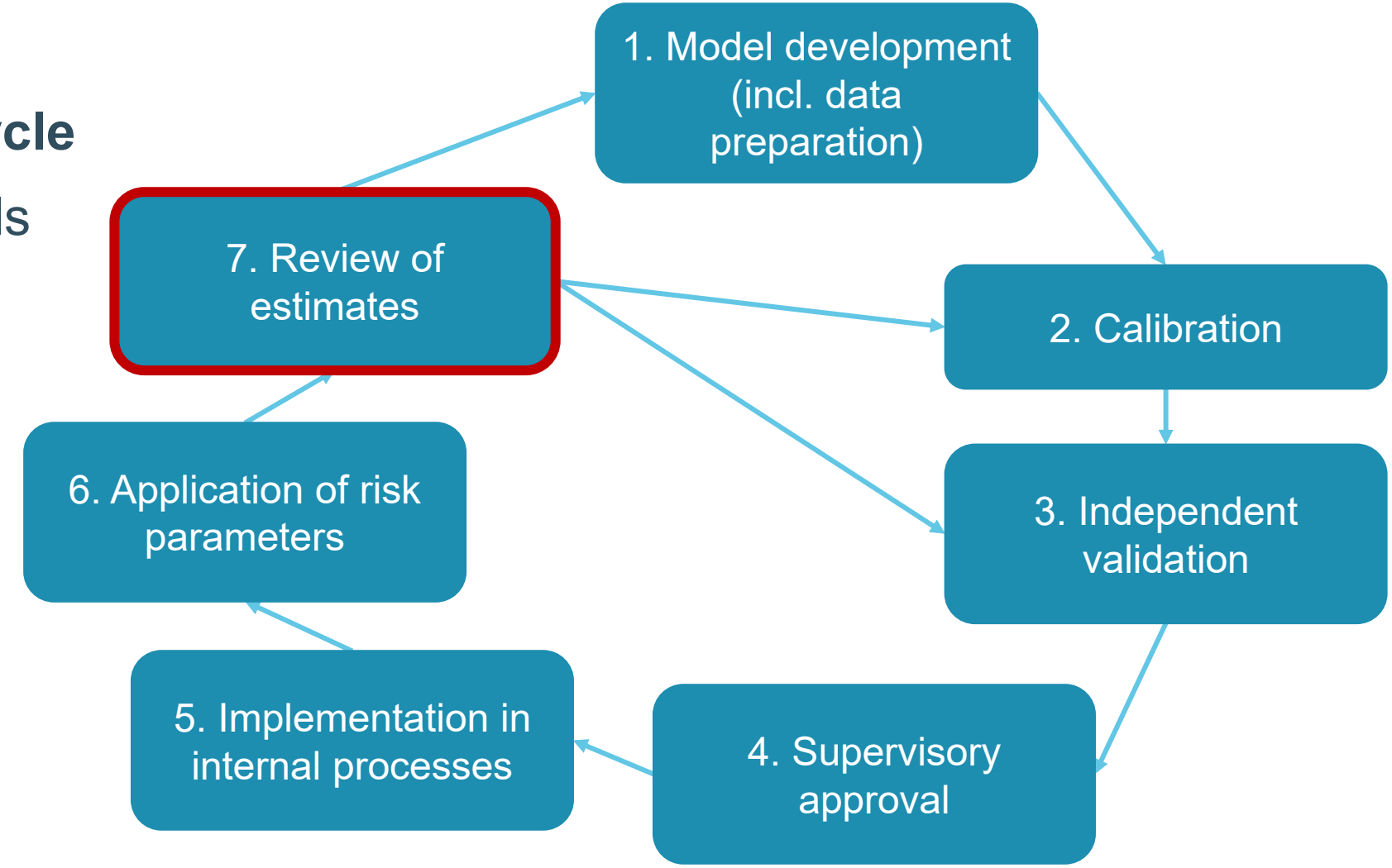
- Link default rates with macro-economy using **time series models**
  - ARIMA, GARCH, … (cf. Chapter 7: Predictive Analytics - Forecasting)
- Key variables:
  - GDP, inflation, unemployment
  - Include lagged effects
- Requires **sufficient historical information**
- E.g.:

$$DR_t = \beta_0 + \beta_1 DR_{t-1} + \beta_2 GDP_{t-1} + \ldots$$



Backtesting and benchmarking

- Credit risk model **lifecycle**
- Validation of IRB models
  - By independent validation team (~auditing)
- Review of estimates
  - Backtesting
  - Benchmarking



Backtesting and benchmarking

- Monitoring performance of models over time
  - Why would model performance change (degrade) over time?

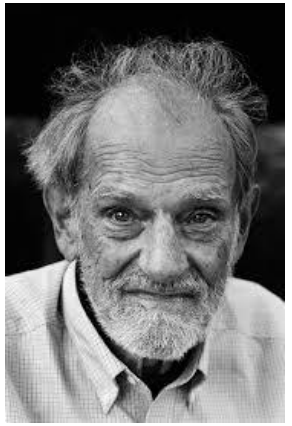
- Common monitoring practices:
  - **Backtesting**:
    - Looking back whether model estimates were accurate
    - Across the levels in model architecture: data (drift?), scorecard (ranking?), ratings (calib.?)
  - **Benchmarking**: comparing operational model's performance with ***baselines***
    - External models (e.g., external ratings, credit scores, etc.)
    - Internal 'challenger' models
      - Other methods
      - Other data
      - …

7. XAI

Feature importance (cf. Chapter 4)

Shapley values

Shapley values



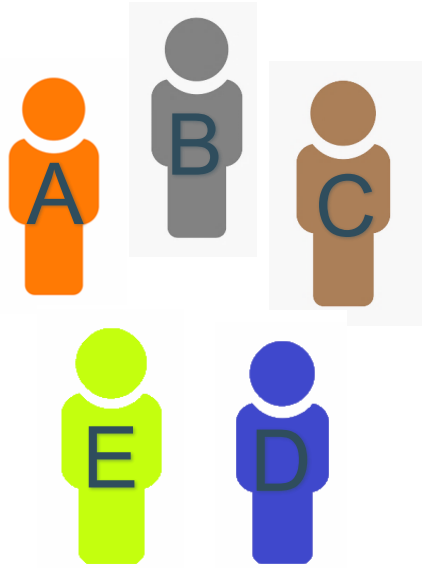
Nobel prize winner Lloyd Shapley

- How to interpret black box model?
  - **XAI:** Shapley values, LIME, counterfactual explanations, …

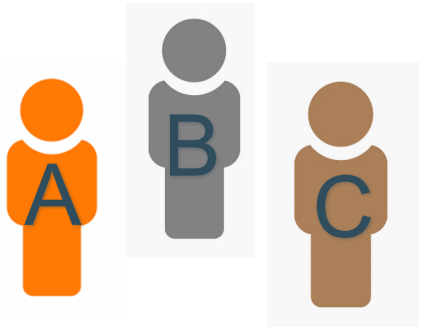
- **Shapley values**
  - Originated in game theory
  - Represents the “payout” for every player in a cooperative game
    - What is the *value* or *contribution* of a player in generating the ‘output’?
    - What is a *correct* payout for different contributors in cooperatively generating value
  - Can be used to measure the contribution of a feature to a model (output)
    - What is the model output with and without a feature’s input?
    - Across all possible cases of other feature’s being available or not

Shapley values

- What is the contribution of a player to a team's performance?
  - We are interested in the **marginal contribution** of each player
  - I.e., the change in output, across all possible *configurations* of the other players participating, or not participating



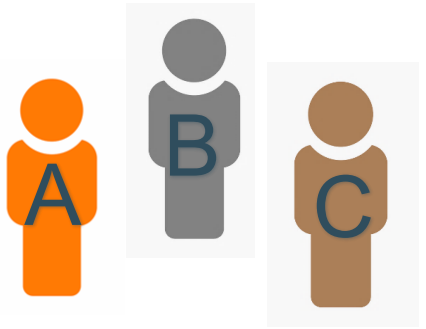
Shapley values



- Suppose the performance, output or **utility** of the team with only player A is 6
  - $v(\{A\}) = 6$, where v is called the **utility or characteristic function**
- Add player B
  - $v(\{A\}) = 6$
  - $v(\{A, B\}) = 10$
  - B has a **marginal contribution** of 4
    - For the specific case where A is already in the team
- Add player C
  - $v(\{A, B, C\}) = 12$
  - C has a **marginal contribution** of 2
    - For the specific case where A and B are already in the team

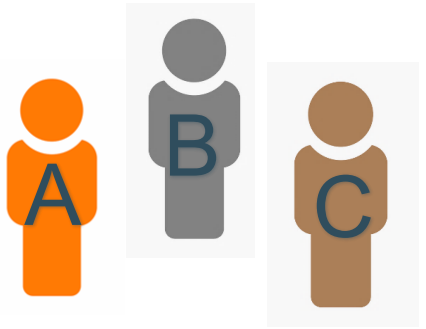
Note: here, A has a **marginal contribution** of 6, for the specific case where no players are in the team yet

Shapley values

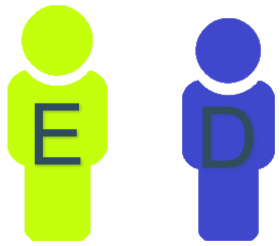


- The marginal contribution typically depends on the players already in the team
  - Add C first to the team
    - $v(\{A\}) = 6$
    - $v(\{A, C\}) = 10$
    - Marginal contribution of C is 4 (vs. 2 when B is already in the team)
  - Add B again
    - $v(\{A, B, C\}) = 12$
    - Marginal contribution of B is 2 (vs. 4 when C is not in the team yet)
- To determine the contribution of C, we need to average her contribution for all possible *formations* of the team
  - I.e., for all possible combinations of players already in the team

Shapley values



- **Note**: contribution of C does not depend on the order in which players were added before adding C:
  - $v(\{A, B, C\}) - v(\{A, B\}) = v(\{B, A, C\}) - v(\{B, A\})$
  - So, we only need to calculate payoff for C once for the set of players selected before adding C, and weigh it with the number of possible combinations of the players selected before
- **Note**: contribution of C does not depend on how you add additional players (D and E) after adding C:
  - So $v(\{A, B, C\}) - v(\{A, B\})$ is same the for sequences ABCDE, ABCED, BACDE and BACED



Shapley values

- The **contribution or Shapley value, φ_j, of a player x_j** is calculated as:

$$\varphi_j = \sum_{S \subseteq \{x_1,\dots,x_p\}\backslash\{x_j\}} \frac{|S|!\,(p - |S| - 1)!}{p!}\Big(v\big(S \cup \{x_j\}\big) - v(S)\Big)$$

- $\{x_1, \dots, x_p\}$ is the set of players in the team, with p the number of players
- $S \subseteq \{x_1, \dots, x_p\}\backslash\{x_j\}$ is a subset S of the players, excluding player x_j
- $\frac{|S|!(p-|S|-1)!}{p!}$ is the **number of possible subsets** with $|S|$ players, and $|S|$ the number of players in S
- $v\big(S \cup \{x_j\}\big) - v(S)$ is the difference in utility when adding player x_j to S

Shapley values

How to evaluate utility v to calculate contribution φ_j?

$$\varphi_j = \sum_{S \subseteq \{x_1,\dots,x_p\} \setminus \{x_j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big(v\big(S \cup \{x_j\}\big) - v(S) \Big)$$

- Using Shapley values to explain predictive models?
- To evaluate the ***importance* of a feature X_j** for a specific observation **x** **(local explanation)**
  - Predicting the outcome of the observation is 'the game'
  - The observed values of the variables are the players that contribute in making the prediction
  - Calculate **contribution φ_j** (Shapley value) of the ***observed value* x_j** of feature X_j
    - i.e., the contribution of 'player' x_j
  - To the prediction of model $\hat{f}$
  - For an observation $\mathrm{x} = \{x_1, \dots, x_j, \dots, x_p\}$
    - i.e., for the 'set of players' $\{x_1, \dots, x_j, \dots, x_p\}$

- To evaluate the ***importance* of a feature X_j** **(global explanation)**
  - **Average the contributions** of the values x_j of a feature X_j
  - To the predictions of model $\hat{f}$
  - Across all observations $\mathrm{x} = \{x_1, \dots, x_j, \dots, x_p\}$ in the data set

Shapley values

How to evaluate utility v to calculate contribution φ_j?

$$\varphi_j = \sum_{S \subseteq \{x_1,\dots,x_p\} \setminus \{x_j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big(v\big(S \cup \{x_j\}\big) - v(S) \Big)$$

- **Utility function v** to evaluate the utility for a **subset of observed values**, $S \subseteq \{x_1, \dots, x_p\}$

$$v(S) = \int \hat{f}(x_1, \dots, x_p) d\mathbb{P}_{x \notin S} - E_X\big[\hat{f}(\mathrm{x})\big]$$

- Example:
  - Three variables in dataset: $X = \{X_1, X_2, X_3\}$
    - E.g., $X = \{age, income, duration\}$
  - An observation is generally defined as: $\mathrm{x} = (x_1, x_2, x_3)$
    - E.g., $\mathrm{x} = (37{,}2700{,}5)$
  - Subset of observed values: $S = (x_1, x_3)$
    - E.g., $\mathrm{S} = (37{,}5)$
  - Utility of S: $v(S) = \int \hat{f}(x_1, x_2, x_3) dx_2 - \hat{f}(\bar{\mathrm{x}})$
    - E.g., for $\mathrm{S} = (37{,}5)$, we may calculate $\int \hat{f}(37, x_2, 5) dx_2$ by averaging $\hat{f}(37, x_2, 5)$ for all observed values of x_2
    - $\hat{f}(\bar{\mathrm{x}})$ is the average estimate for all observations in the dataset = the estimate for the average observation

Shapley values

How to evaluate utility v to calculate contribution φ_j?

$$\varphi_j = \sum_{S \subseteq \{x_1,\dots,x_p\} \setminus \{x_j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big(v\big(S \cup \{x_j\}\big) - v(S) \Big)$$

- **Utility function** to evaluate the utility for a **subset of observed values**, $S \subseteq \{x_1, \dots, x_p\}$

$$v(S) = \int \hat{f}(x_1, \dots, x_p) d\mathbb{P}_{x \notin S} - E_X\big[\hat{f}(\mathrm{x})\big]$$

- To obtain the estimate of the model over the subset of observed values, you have to *integrate out* the other values that are not in the subset
  - Integrate out? → Average estimate for all possible combinations of values for the variables that are not in the subset
- The contribution is then the difference of this prediction from **the average prediction** $E_X\big[\hat{f}(\mathrm{x})\big]$
  - The average value can be approximated with the mean value of your target in the data
  - Sometimes also called an intercept model.
- Note: if you subtract two utilities for different subsets (cf. φ_j), this baseline term disappears

Shapley values

- Calculation Shapley values: **computationally expensive**!
  - Evaluating the utility function requires the evaluation of model estimates across all possible combinations of values of the variables not in the subset ('*integrate out*')
  - Calculating the contribution requires the calculation of utility for all subsamples
  - Exponential increase in # of subsets for increasing # of features!
- In practice, Shapley values are often **approximated** using a Monte Carlo sampling approach:

$$\hat{\varphi}_j = \frac{1}{M} \sum_{m=1}^{M} \left(\hat{f}\left(x_{+j}^m\right) - \hat{f}\left(x_{-j}^m\right) \right)$$

  - For every feature j, **a random permutation of other features is sampled M times**
  - The difference in predicted value with and without feature j is calculated, $\hat{f}\left(x_{+j}^m\right) - \hat{f}\left(x_{-j}^m\right)$
  - Shapley value is then calculated as the average across these samples

Shapley values

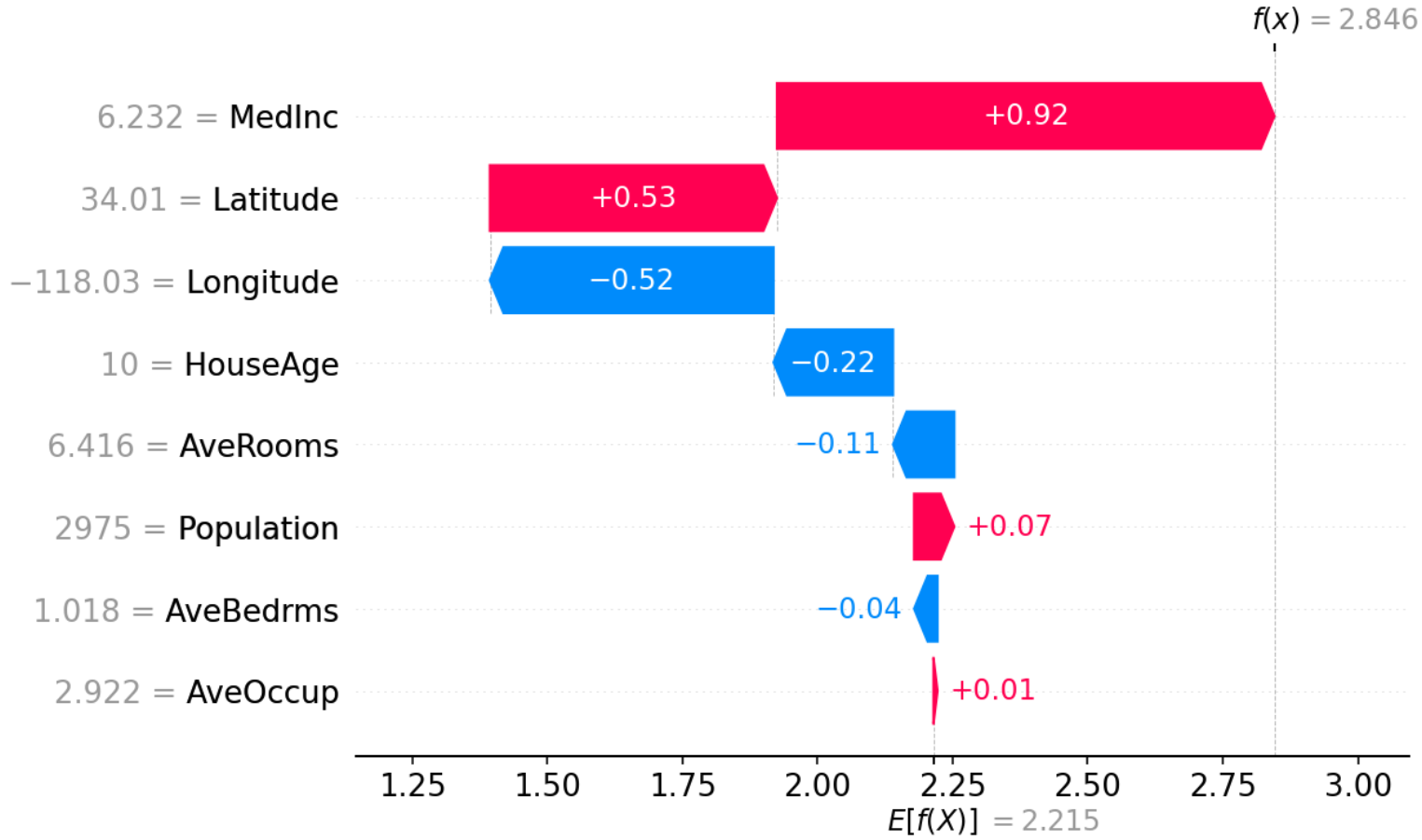
- Assume observation x (= Bart) has features: age, income, R, F, M, debt
- Say we want to determine importance (contribution, Shapley value) of F for Bart
- For m = 1,…,N
  - Draw **random instance** z from data (e.g., z = Tim)
  - Choose **random permutation of features** (e.g., age, M, income)
  - Construct two new instances
  - $\varphi_j^m = \hat{f}(x_{+j}) - \hat{f}(x_{-j})$
- Finally, calculate the contribution of feature F for Bart as: $\varphi_j(x) = \frac{1}{N}\sum_{n=1}^{N} \varphi_j^n$

	R	Debt	F	age	M	income
Bart (= x)	5	100	2	44	1000	2500
Tim (= z)	10	50	4	36	500	3000

	R	Debt	F	age	M	income
Bart (x_{+j})	5	100	2	36	500	3000
Bart (x_{-j})	5	100	4	36	500	3000

Shapley values

- Visualizing Shapley values:
  - Waterfall plots
    - Local explanation



Shapley values

- Visualizing Shapley values:
  - Waterfall plots
    - Local explanation
  - Beeswarm plots
    - Global view

      … on local explanations



Q&A